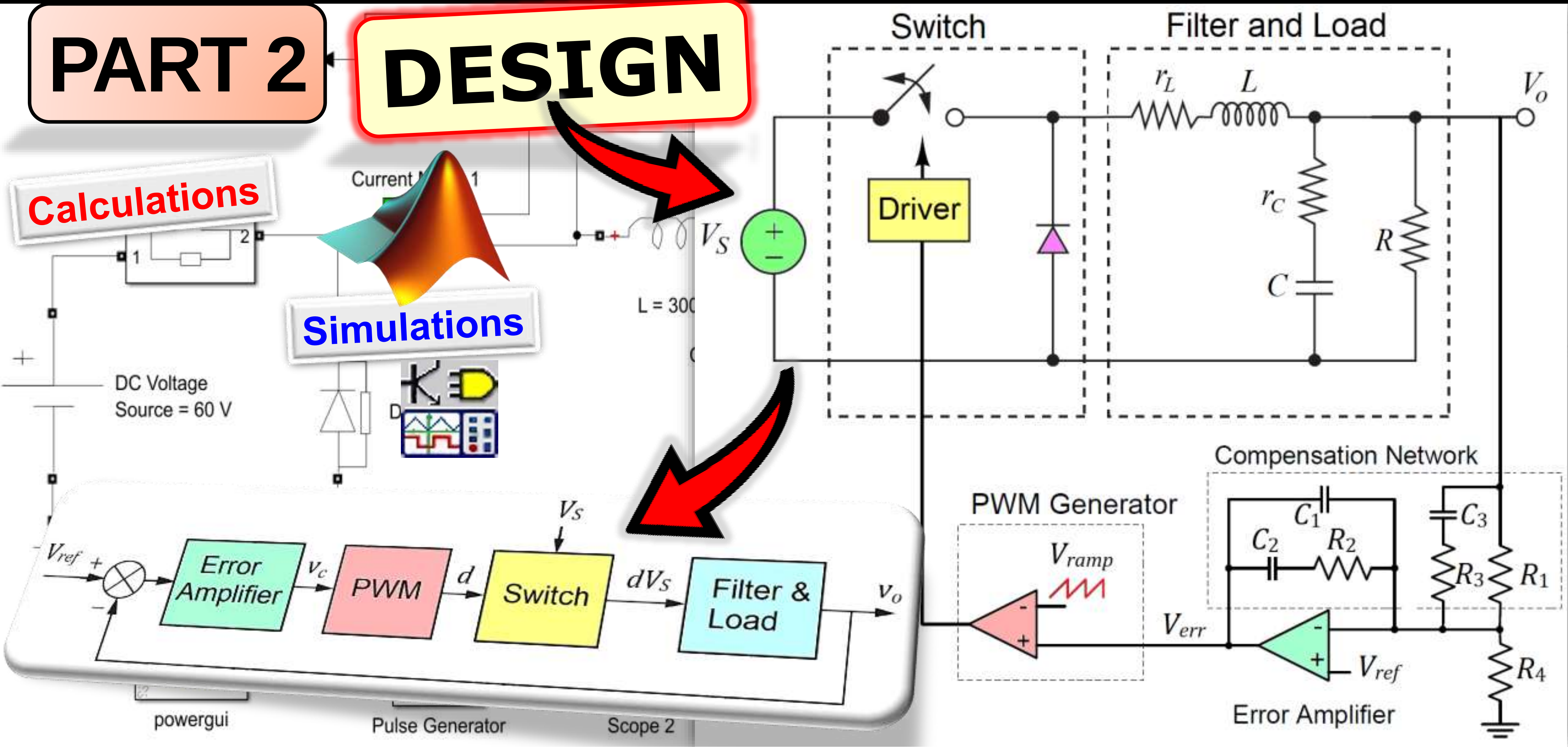
BUCK CONVERTER



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  2. Specifications
  3. Component Values
  4. Component Ratings

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  1. Simulations
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     b. MATLAB/Simulink
  2. Fine Tuning

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  2. Frequency Response Analysis
  3. Root Locus Analysis
  4. Controller Circuits

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  3. Compensator Values - *Pole Placement Method*
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  1. Simulations
     a. TINA-TI Spice
     b. MATLAB/Simulink
  2. Fine Tuning

PART 2A: Theory

1. Time Response Analysis

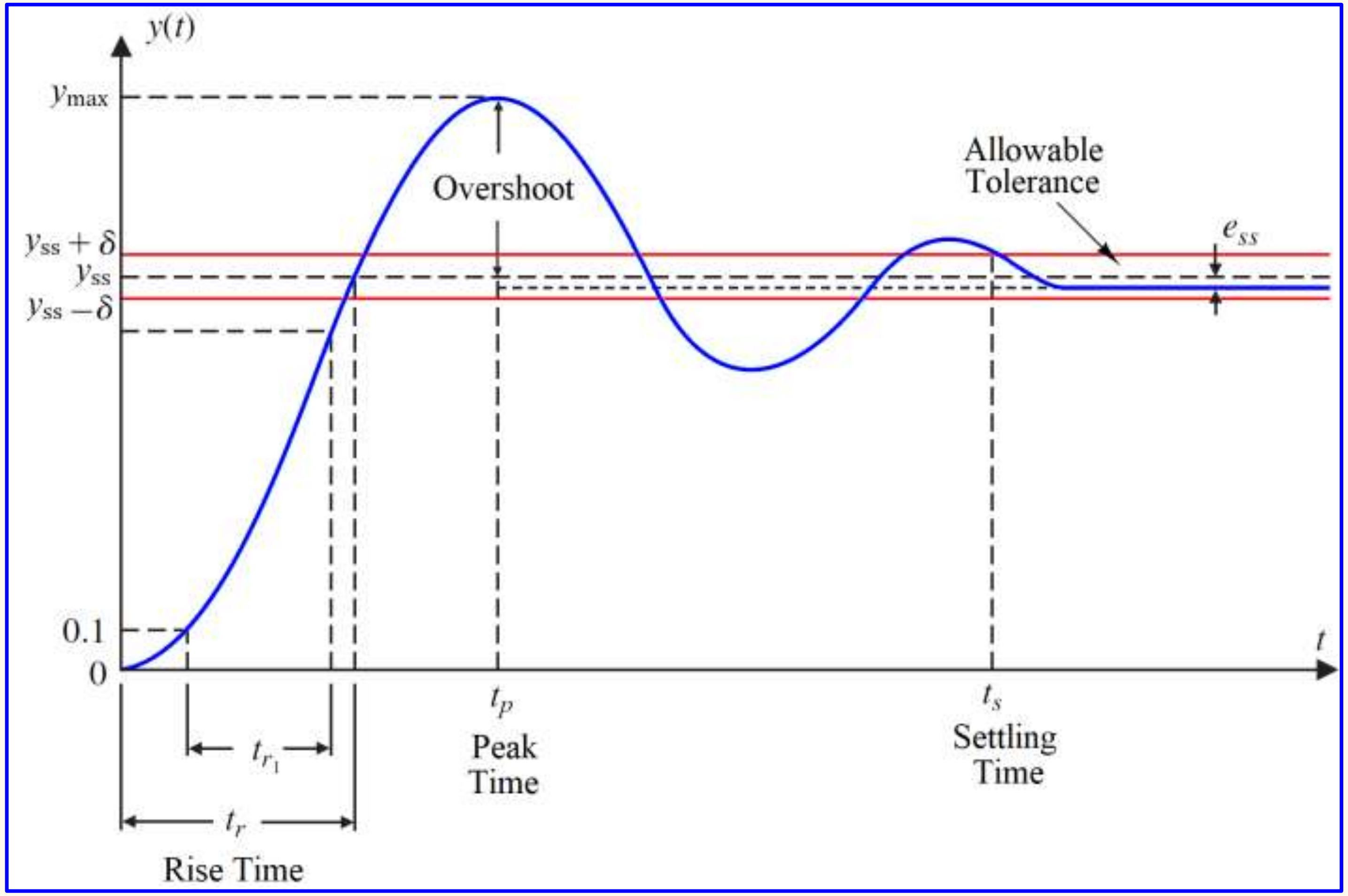
❖ **Unit-Step Response *(Second-Order System)***

$$y(t) = 1 - e^{-\omega_n \zeta t}\left(\cos\left(\omega_n\sqrt{1-\zeta^2}\,t\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}\,t\right)\right)$$

$$y(t) = 1 - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\omega_n \zeta t}\cos\left(\omega_n\sqrt{1-\zeta^2}\,t - \phi\right)$$

$$y(t) = 1 - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\omega_n \zeta t}\cos(\omega_d t - \phi)$$

with $\phi = \arctan\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$



- **General Second-Order System Model**

$$\boxed{H(s) = \frac{K_{DC}\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

- **Natural Frequency:** ω_n
  Frequency of **oscillation** of the system **without damping**

- **Damping Ratio:** ζ $\quad \zeta = \dfrac{\text{Exponential decay frequency}}{\text{Natural frequency}}$

- **DC Gain:** K_{DC}
  Steady-state value (DC gain) of the system

NOTE: this model only holds for an **all-pole second-order system**, thus **without additional poles, zeros, delays or limitations**. If the system is **not a pure all-pole second-order system**, the response can ***deviate*** from the **calculated response**. However, we can use this second-order model to ***estimate*** the performance of other and **higher order systems**.

1. Time Response Analysis

- **Peak Time:** **time required** to **reach** the **first peak value**. Peak time is denoted as t_p
  The peak time is found by differentiating $y(t)$ and setting $dy(t)/dt$ equal to zero.

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

- **Overshoot:** The **amount** that the waveform **overshoots** the **steady-state value** (final value) at the peak time. Overshoot is denoted as M_p.

$$M_p = \frac{y_{\max} - y_{ss}}{y_{ss}} \qquad y_{\max} = y(t_p) = 1 + e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

For unit step input, $y_{ss} = 1$

Then, we get $M_p = \frac{1 + e^{-\pi\zeta/\sqrt{1-\zeta^2}} - 1}{1} = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ or $\zeta = \sqrt{\frac{\ln^2(M_p)}{\ln^2(M_p) + \pi^2}}$

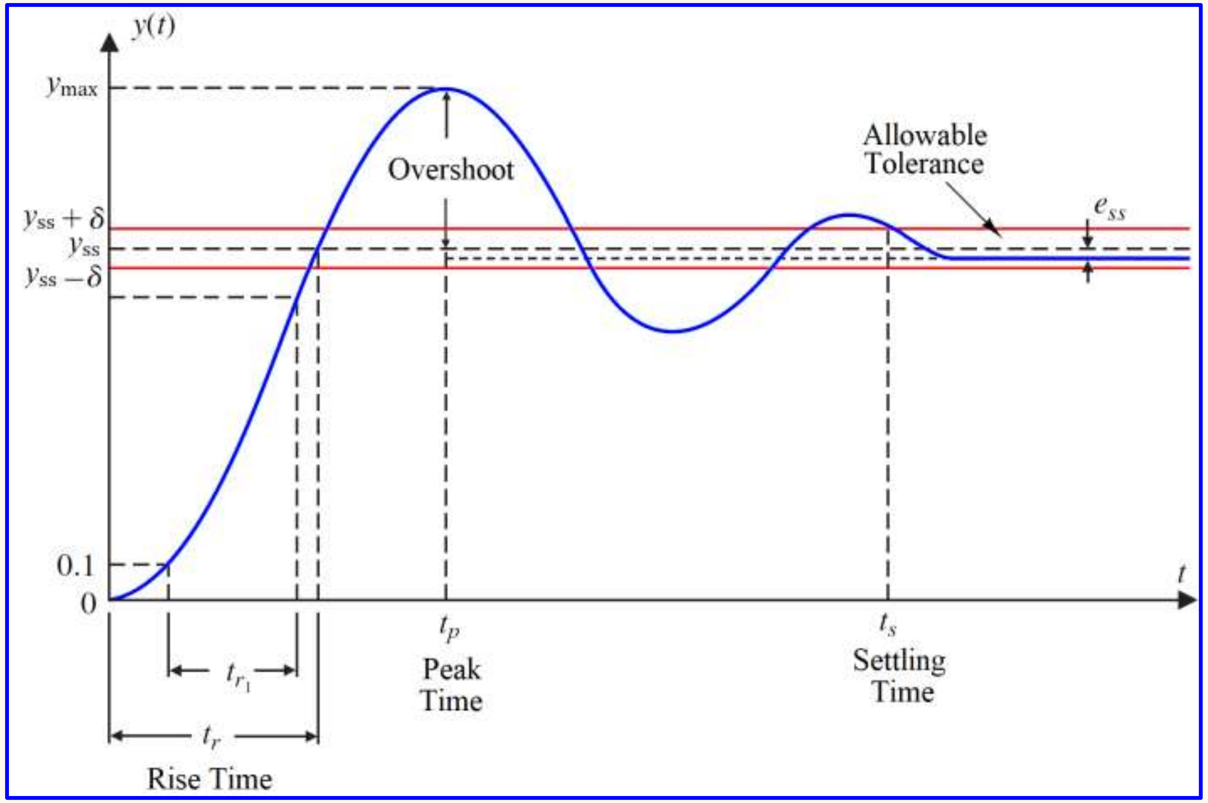
- **Rise Time:** The **time** for the waveform **to go** from 0% to 100% of its **final value**.

Exact expression: $t_r = \frac{\arctan\left(\sqrt{1-\zeta^2}/\zeta\right)}{\omega_n\sqrt{1-\zeta^2}}$

Approximated expression: $t_r = \frac{2.16\zeta + 0.6}{\omega_n}$

- **Settling Time:** Time for the response to **reach** and **stay within a specific percentage** of the **steady-state value**.

➢ Settling time with $\pm 2\%$ **criterion** is called ***indication time*** and is denoted as $t_s(2\%)$
➢ Settling time with $\pm 5\%$ **criterion** is called ***response time*** and is denoted as $t_s(5\%)$



Indication time: $M_p = e^{-\pi\zeta t/\sqrt{1-\zeta^2}} = 0.02 \rightarrow t = -\frac{\ln\left(0.02\sqrt{1-\zeta^2}\right)}{\zeta\omega_n} \equiv t_s(2\%)$

Approximately: $t_s(2\%) = \frac{4}{\zeta\omega_n}$

Response time: $M_p = e^{-\pi\zeta t/\sqrt{1-\zeta^2}} = 0.05 \rightarrow t = -\frac{\ln\left(0.05\sqrt{1-\zeta^2}\right)}{\zeta\omega_n} \equiv t_s(5\%)$

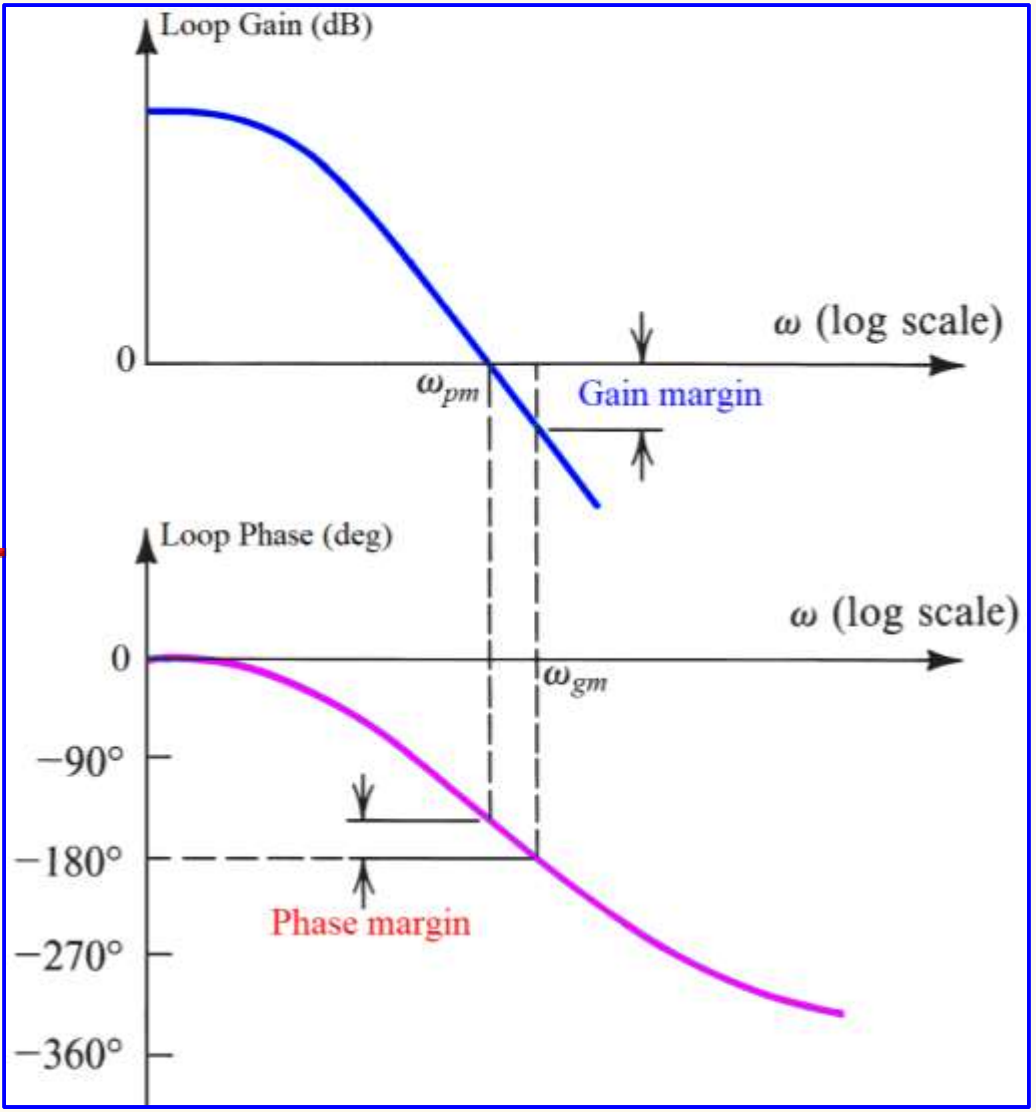
Approximately: $t_s(5\%) = \frac{3}{\zeta\omega_n}$

2. Frequency Response Analysis

❖ Stability Margins

- **Gain Margin:**
  1. Find the **frequency** where the **loop phase** is $-180°$. This frequency is called ω_{gm}.
  2. Determine the **loop gain** at ω_{gm}.
  3. The **gain margin (GM)** is the gain required to ***raise*** the **magnitude curve** to 0 dB.

- **Phase Margin:**
  1. Find the **frequency** where the **loop gain** is 0 dB. This frequency is called ω_{pm}.
  2. The **phase margin (PM)** is the **difference** between the **loop phase** at ω_{pm} and $-180°$.



Example:

Loop transfer function: $L(j\omega) = \dfrac{1000}{(j\omega+4)(j\omega+8)(j\omega+10)}$

Determine:

- Phase margin (*PM*)
- Gain margin (*GM*)

Phase Margin: $|L(j\omega)| = 1 \;\Rightarrow\; \dfrac{1000}{\sqrt{\omega^2+4^2}\sqrt{\omega^2+8^2}\sqrt{\omega^2+10^2}} = 1$

Solving:
$\omega = 6.79 \text{ rad/s} \equiv \omega_{pm}$

$$\arg(\omega_{pm}) = -\arctan\left(\frac{\omega_{pm}}{4}\right) - \arctan\left(\frac{\omega_{pm}}{8}\right) - \arctan\left(\frac{\omega_{pm}}{10}\right) = -134°$$

$$PM = 180 + \arg(\omega_{pm}) = 46°$$

Gain Margin: $\arg(\omega) = -\arctan\left(\frac{\omega}{4}\right) - \arctan\left(\frac{\omega}{8}\right) - \arctan\left(\frac{\omega}{10}\right) = -180°$

Solving:
$\omega = 12.33 \text{ rad/s} \equiv \omega_{gm}$

$$|L(j\omega_{gm})| = \frac{1000}{\sqrt{\omega_{gm}^2+4^2}\sqrt{\omega_{gm}^2+8^2}\sqrt{\omega_{gm}^2+10^2}} = 0.3307\ (-9.61\text{ dB})$$

$$GM = \frac{1}{|L(j\omega_{gm})|} = 3.024\ (9.61\text{ dB})$$

2. Frequency Response Analysis

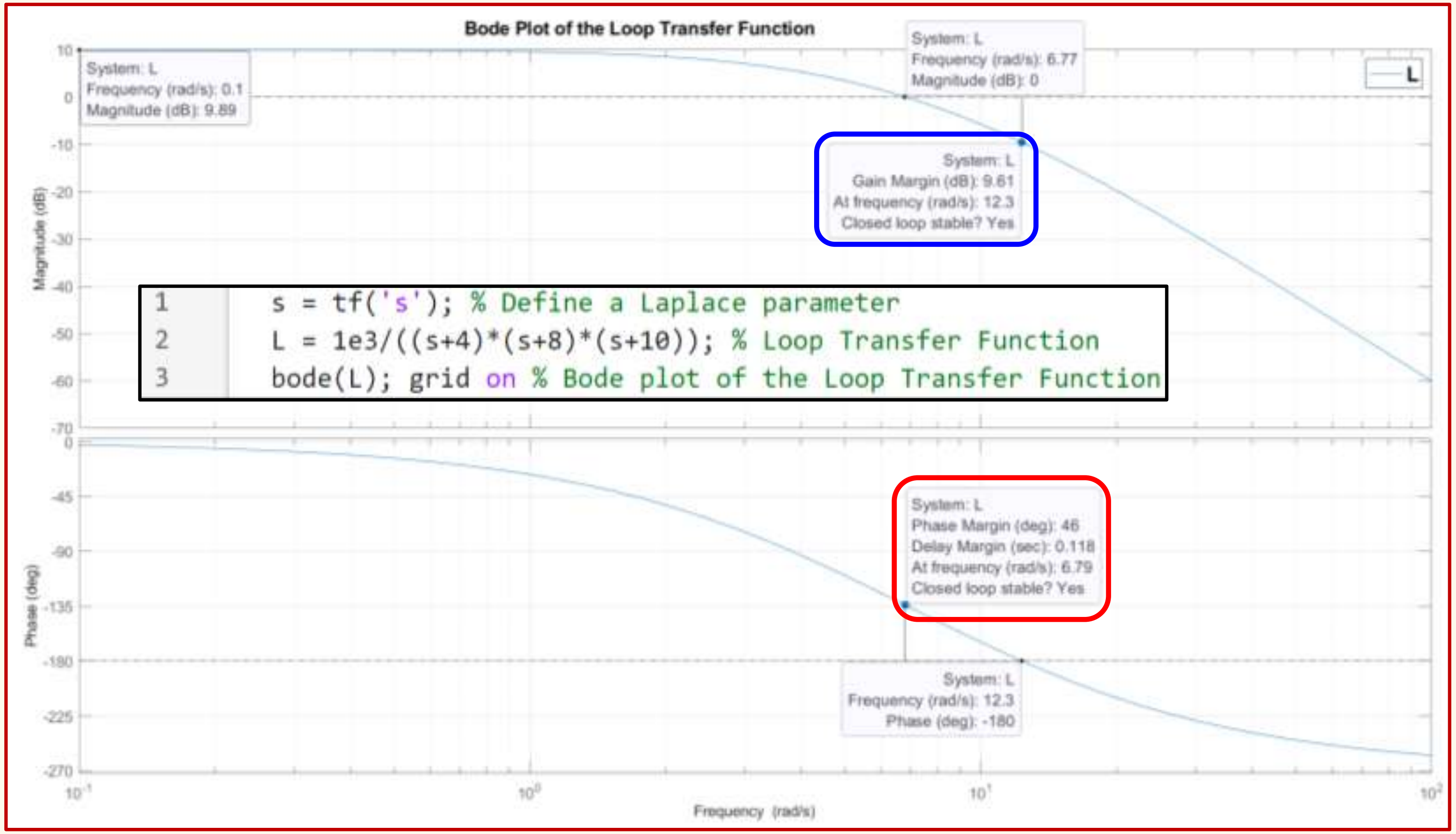
Example:

Loop transfer function: $L(j\omega) = \dfrac{1000}{(j\omega+4)(j\omega+8)(j\omega+10)}$

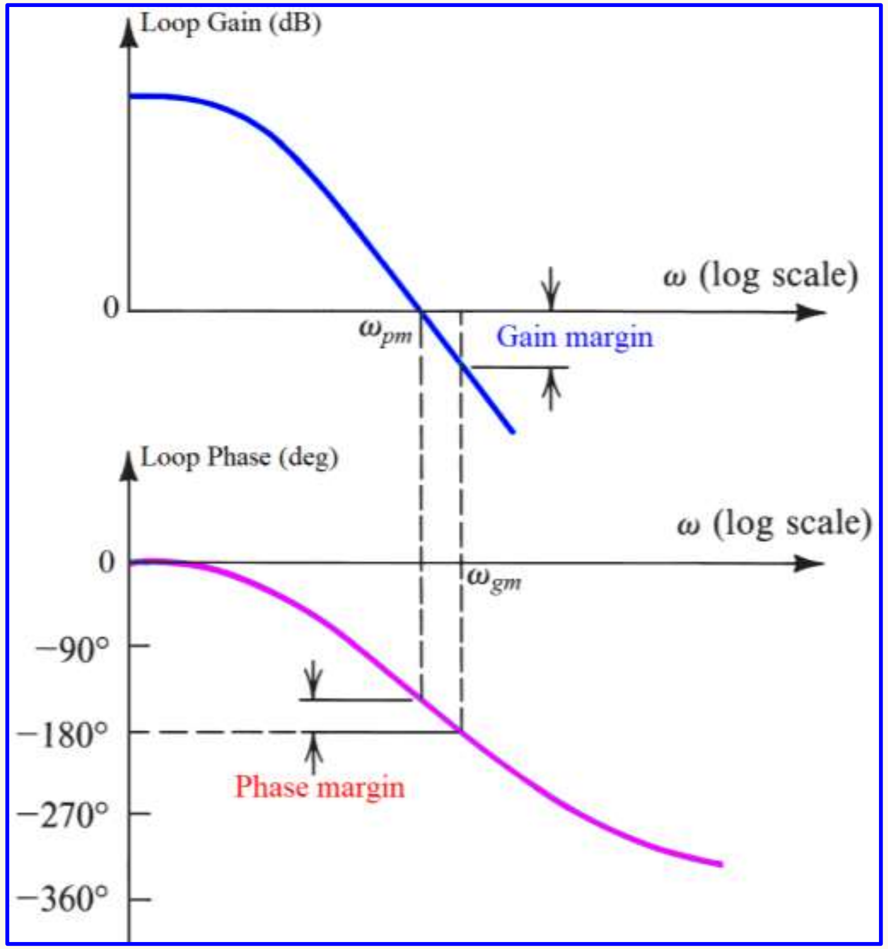
Determine:

- Phase margin (*PM*)
- Gain margin (*GM*)

Simulation Results:



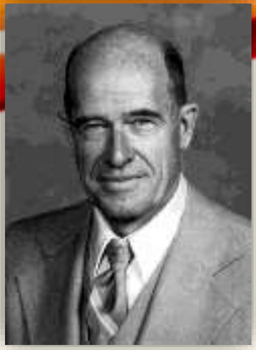

```
s = tf('s'); % Define a Laplace parameter
L = 1e3/((s+4)*(s+8)*(s+10)); % Loop Transfer Function
bode(L); grid on % Bode plot of the Loop Transfer Function
```



$\omega_{pm} = 6.79\ \mathrm{rad/s} \quad PM = 180 + \arg(\omega_{pm}) = 46°$

$\omega_{gm} = 12.33\ \mathrm{rad/s} \quad GM = \dfrac{1}{|L(j\omega_{gm})|} = 3.024\ (9.61\ \mathrm{dB})$

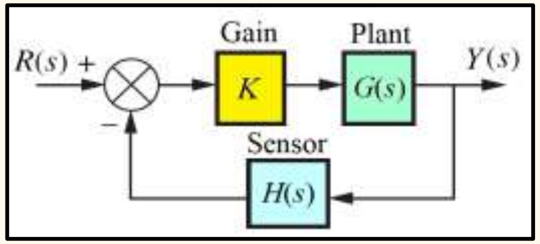
3. Root Locus Analysis



❖ Root Locus Method

- Analysis and design for **stability** and **transient response** (*W.R. Evans,* 1948).
- **Graphical presentation** of the **closed-loop poles** as a **function** of a **system parameter**.
- Display the effects on ***overshoot***, ***settling time***, ***rise time*** and ***peak time***.

Example:



$$G(s) = \frac{2}{s-1} \qquad H(s) = \frac{1}{s+3}$$

Determine the range of stability.

Root Locus Equation: $1 + L(s) = 0$

$$1 + KG(s)H(s) = 0 \Rightarrow 1 + \frac{2K}{(s-1)(s+3)} = 0$$

$$(s-1)(s+3) + 2K = 0$$

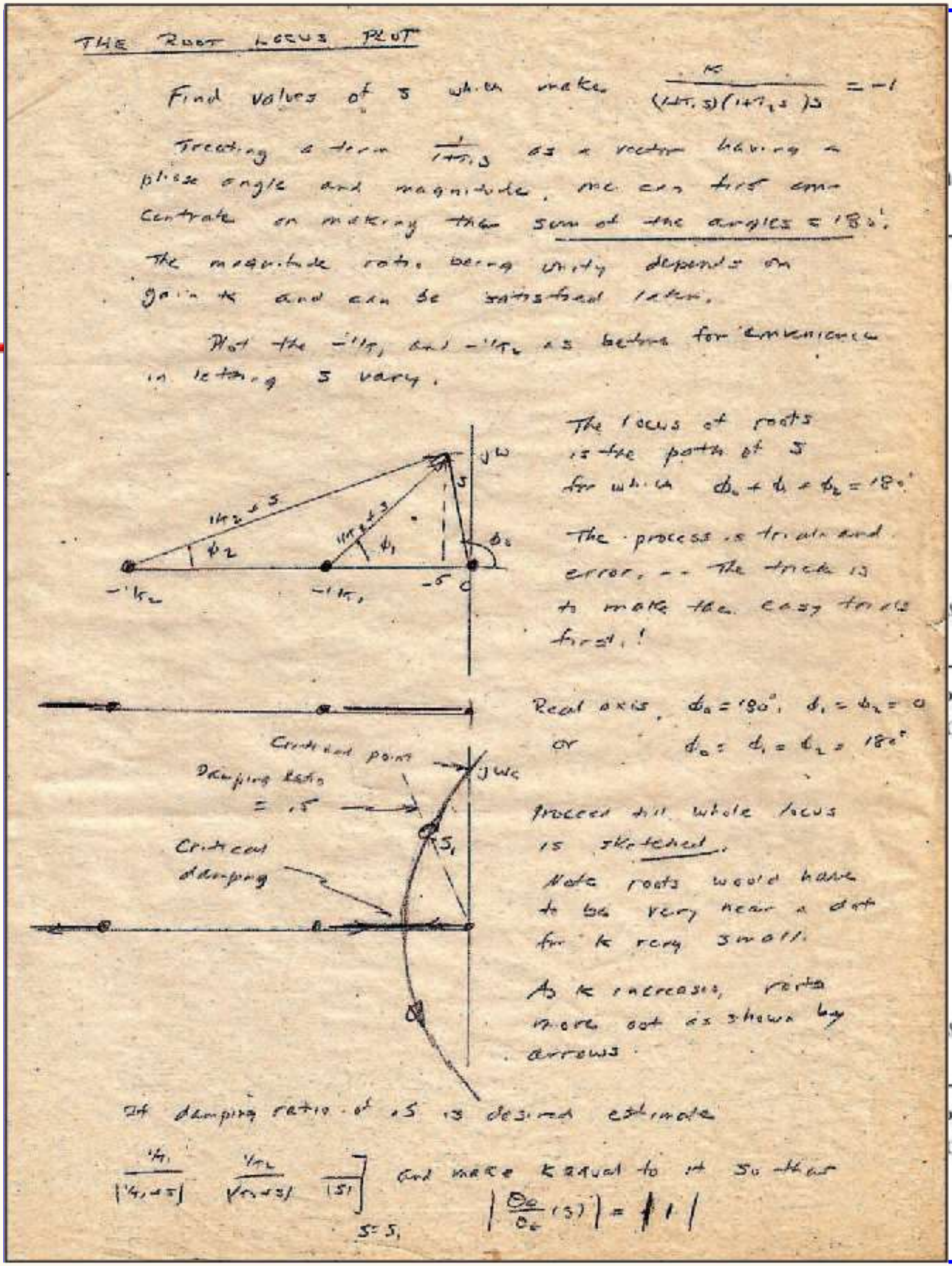
$$s^2 + 2s - 3 + 2K = 0$$

When $-3 + 2K = 0$, we have: $s^2 + 2s = 0$

$$s(s+2) = 0$$

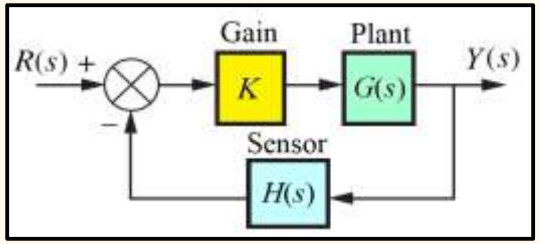
$$-3 + 2K \geq 0 \Rightarrow K \geq 1.5$$

Stability Range



3. Root Locus Analysis

Example:



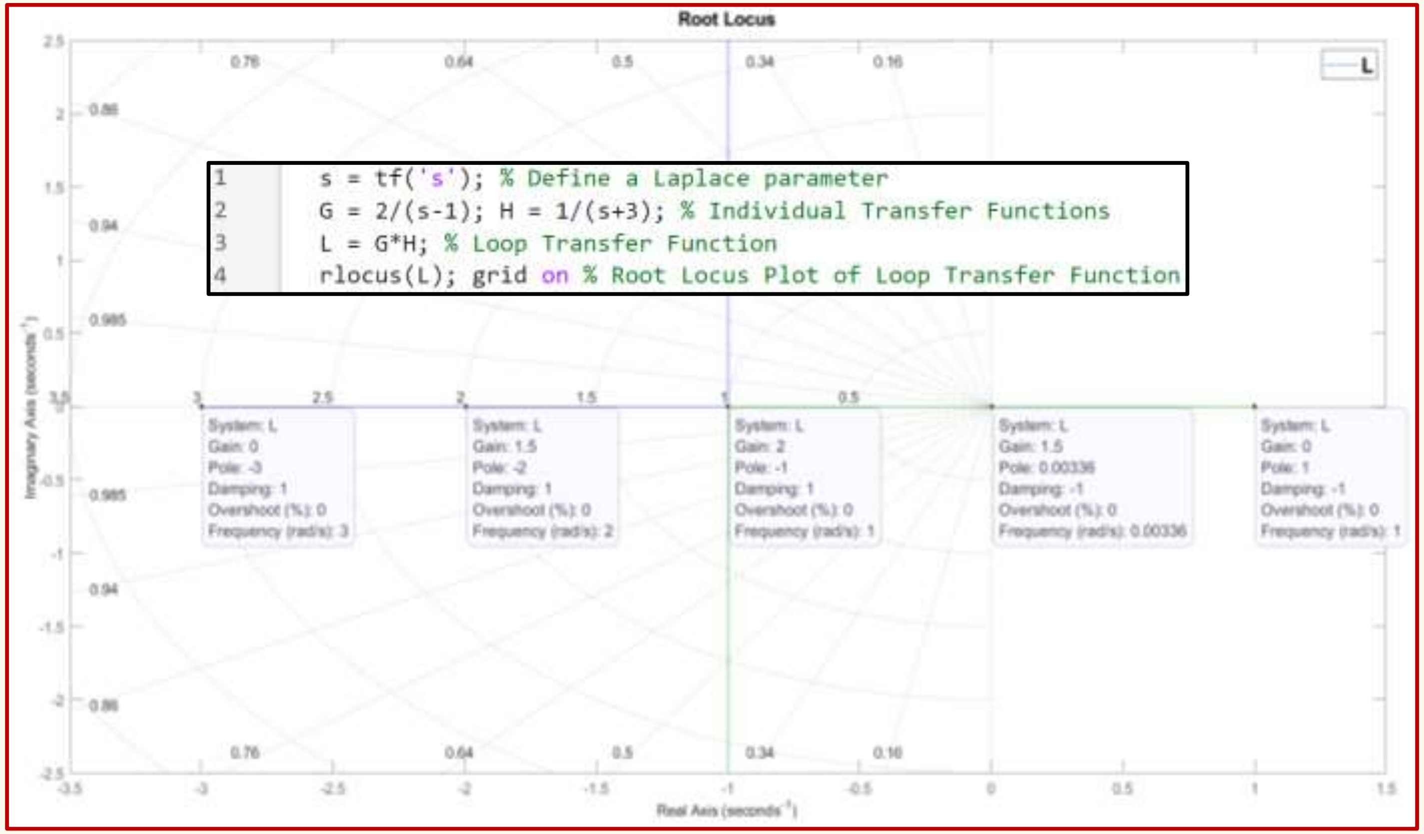
$$G(s) = \frac{2}{s-1} \qquad H(s) = \frac{1}{s+3}$$

Determine the range of stability.

$$s^2 + 2s - 3 + 2K = 0 \qquad K \geq 1.5$$

Stability Range

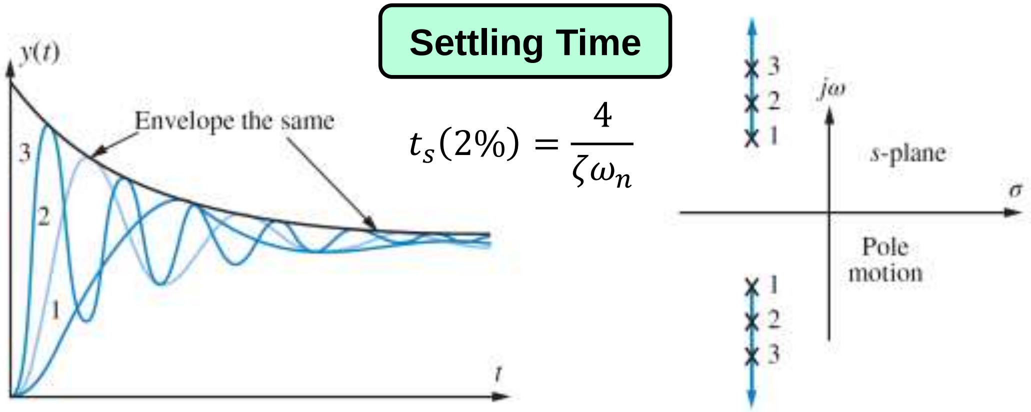
Simulation Results:




```
s = tf('s'); % Define a Laplace parameter
G = 2/(s-1); H = 1/(s+3); % Individual Transfer Functions
L = G*H; % Loop Transfer Function
rlocus(L); grid on % Root Locus Plot of Loop Transfer Function
```

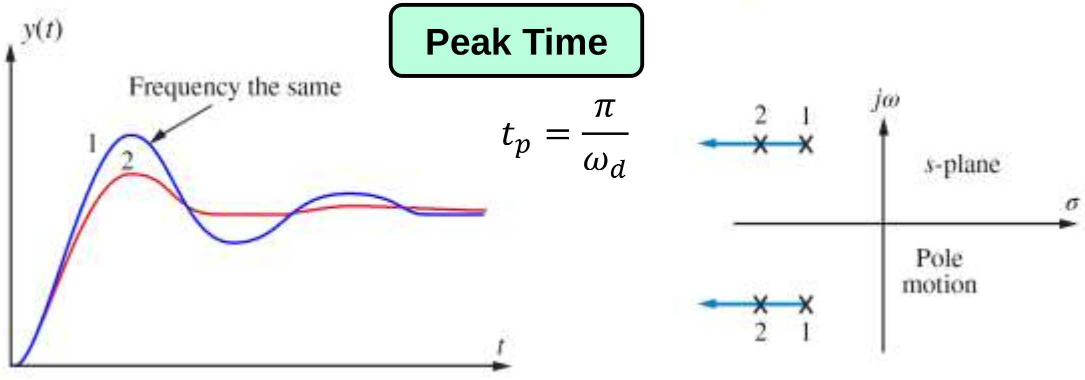
Settling Time

$$t_s(2\%) = \frac{4}{\zeta\omega_n}$$



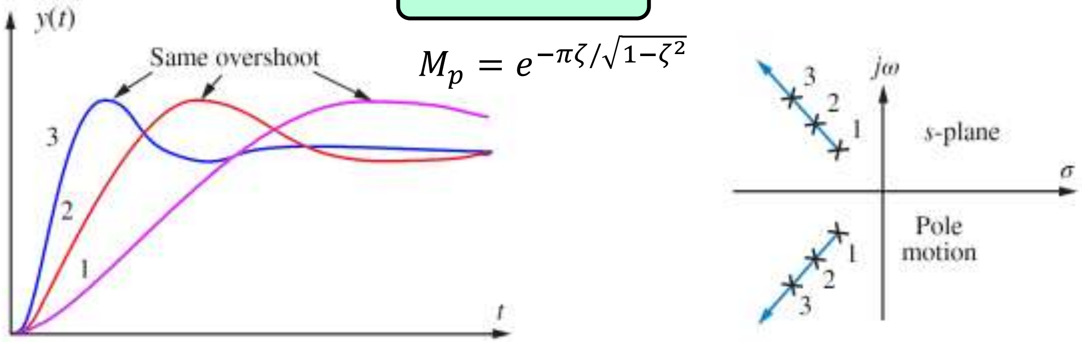
Peak Time

$$t_p = \frac{\pi}{\omega_d}$$



Overshoot

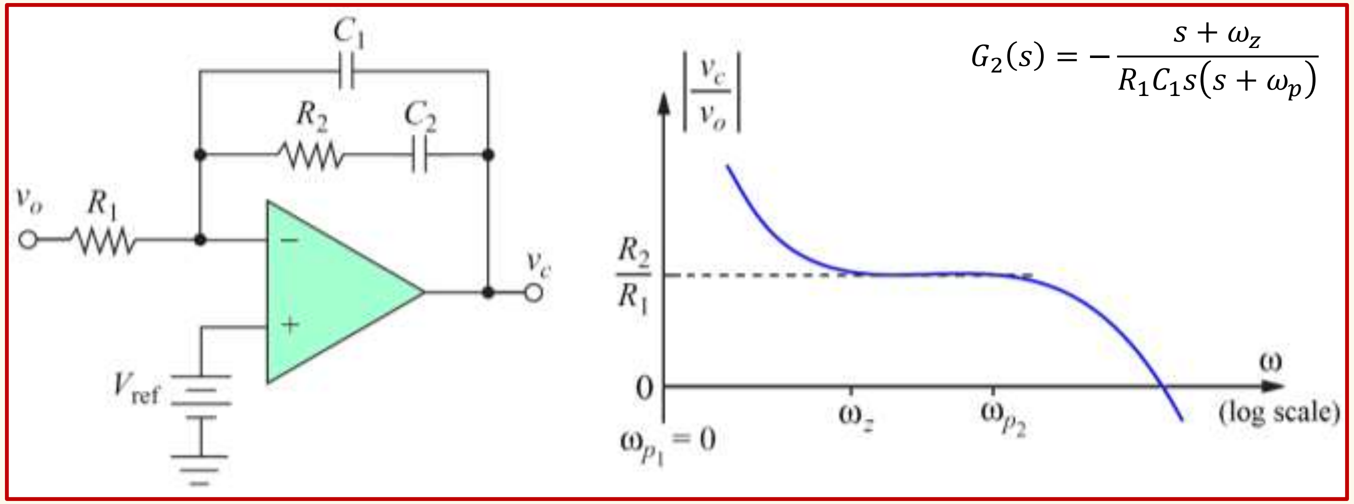
$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$



4. Controller Circuits

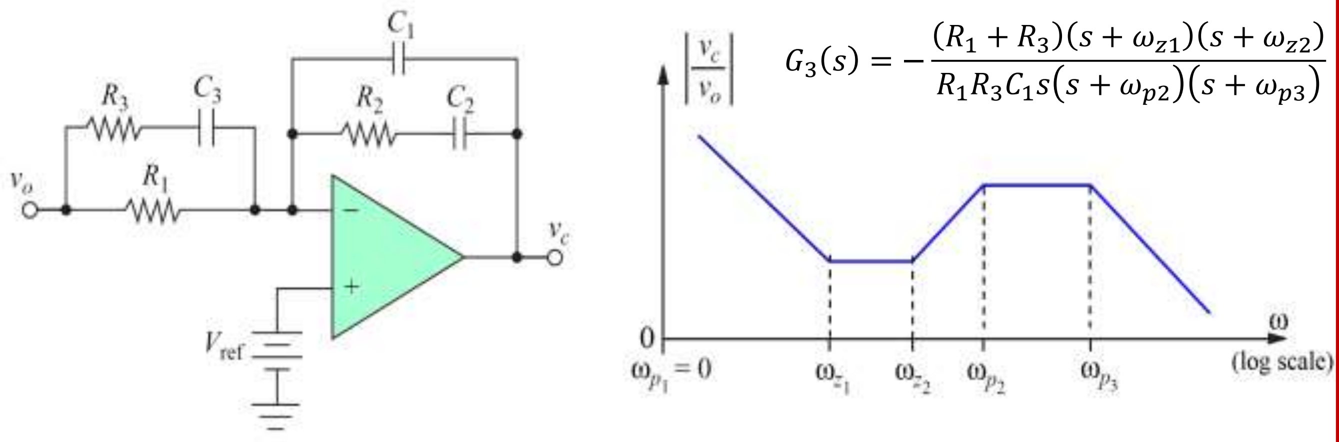
❖ **Type 2 Compensated Error Amplifier**

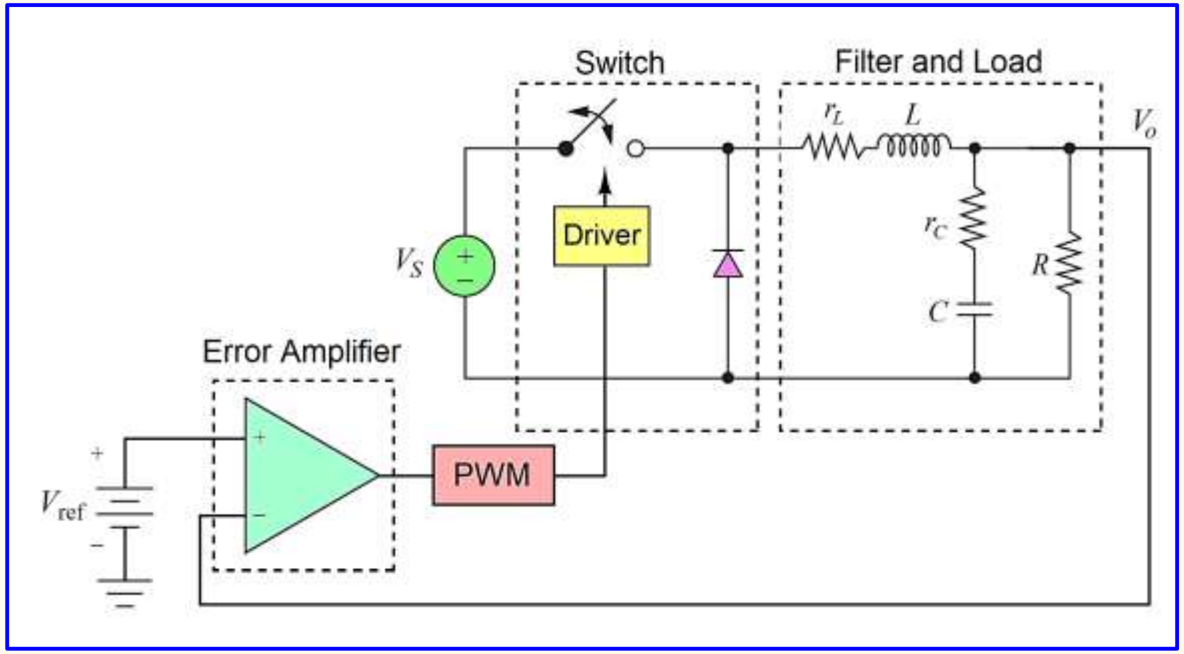
$$G_2(s) = -\frac{s+\omega_z}{R_1 C_1 s(s+\omega_p)}$$

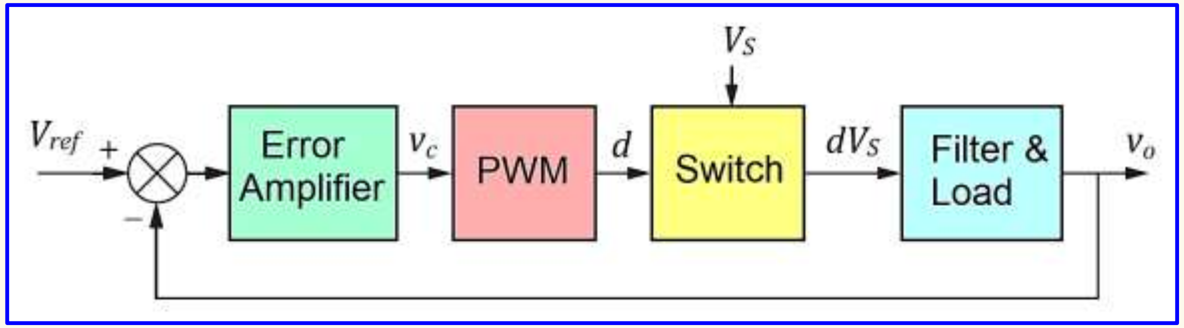


❖ **Type 3 Compensated Error Amplifier**

$$G_3(s) = -\frac{(R_1+R_3)(s+\omega_{z1})(s+\omega_{z2})}{R_1 R_3 C_1 s(s+\omega_{p2})(s+\omega_{p3})}$$

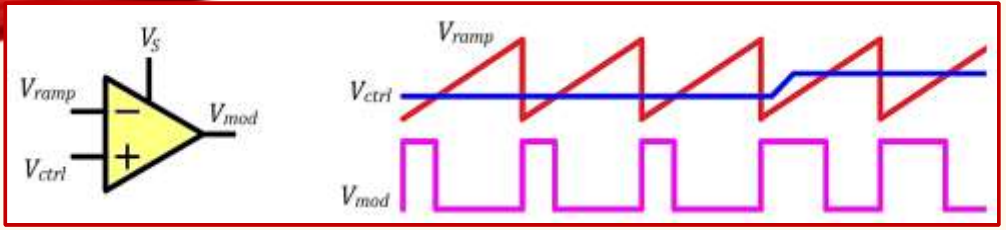






4. Controller Circuits

❖ **Modulator:**



Modulator Average Output Voltage: $V_{mod} = \frac{1}{T}\int_0^T v_{mod}(t)dt = \frac{1}{T}\int_0^{t_{on}} V_S dt$

Modulator Transfer Function: $M(s) = \frac{\Delta V_{mod}}{\Delta V_{ctrl}}$

$$M(s) = \frac{dV_{mod}}{dV_{ctrl}} = \frac{d}{dV_{ctrl}}\left(V_S\frac{V_{ctrl}}{V_{ramp}}\right) \qquad \boxed{M(s) = \frac{V_S}{V_{ramp}}}$$

❖ **Filter & Load:**

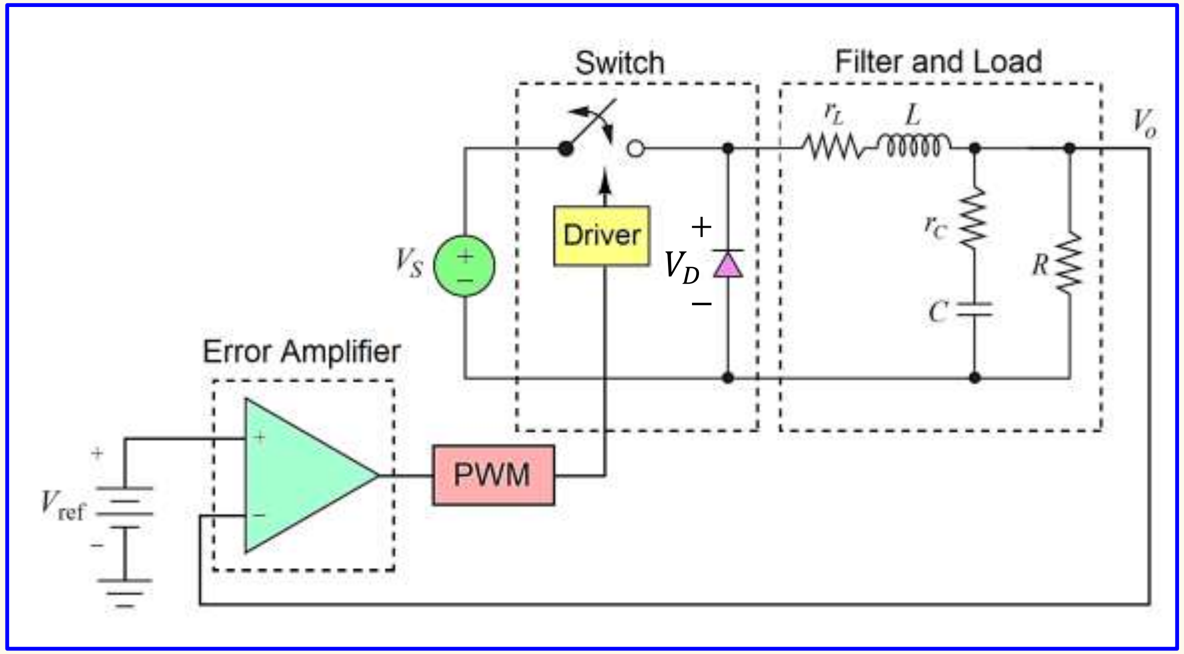
Filter & Load Transfer Function: $F(s) = \frac{V_o}{V_D} = \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right)LCs^2}$

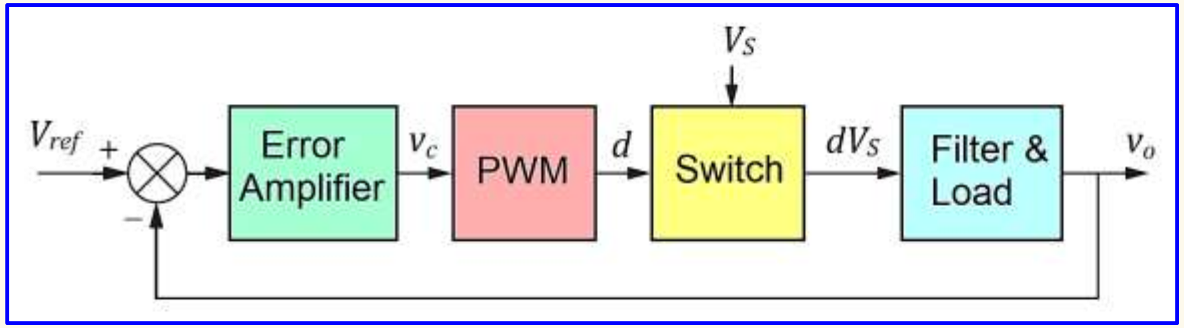
❖ **Compensator:**

Compensator Transfer Function: $G(s) = \frac{V_c}{V_o} = \frac{A_{OL}(s)}{1 + A_{OL}(s)\beta(s)} \approx \frac{A_{OL}(s)}{A_{OL}(s)\beta(s)} \approx \frac{1}{\beta(s)}$

Error Amplifier Open-Loop Transfer Function: $A_{OL}(s) = \frac{A_{DC}}{1 + \frac{s}{\omega_{pe}}}$ $\qquad A_{OL}(s)\beta(s) \gg 1$

Feedback Transfer Function: $\beta(s)$



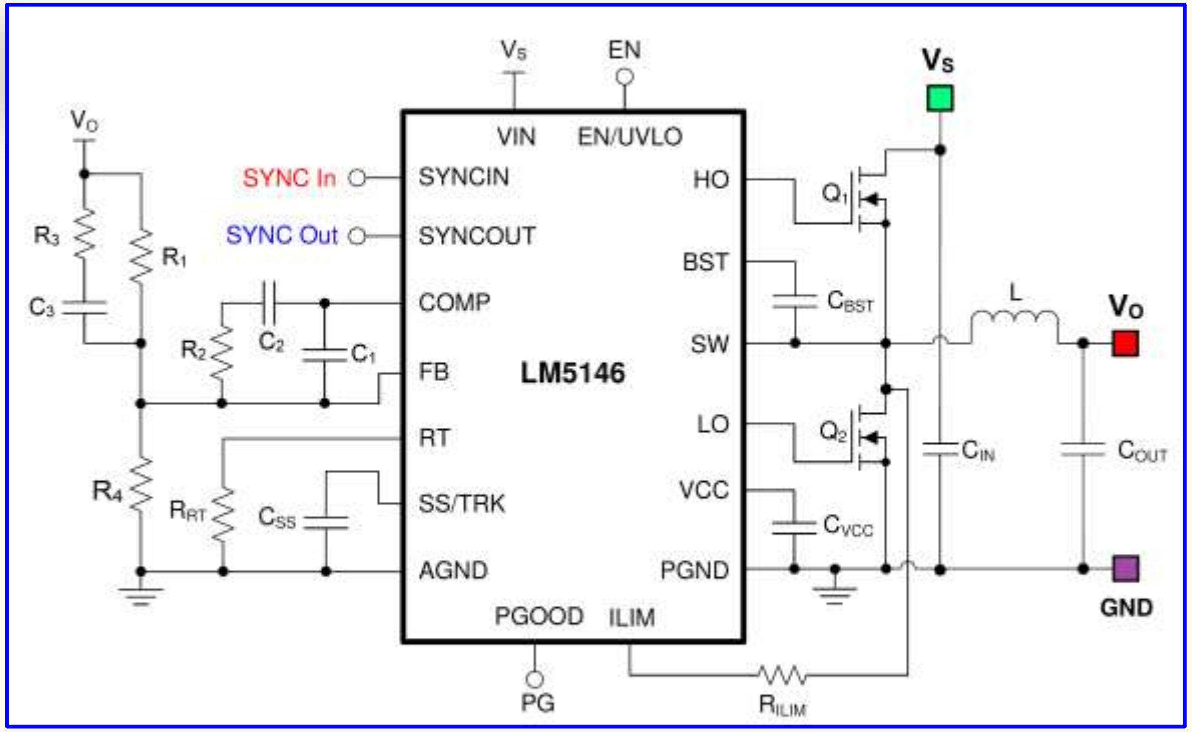


PART 2B:
CALCULATIONS

❖ Specifications:

Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60$ V
- Average output voltage $V_o = 15$ V
- Average output current $I_o = 2$ A
- Maximum percentage peak-peak output ripple voltage $\Delta V_o / V_o = 0.01\ (1\%)$



❖ Buck Converter Component Values & Selections:

PART 1

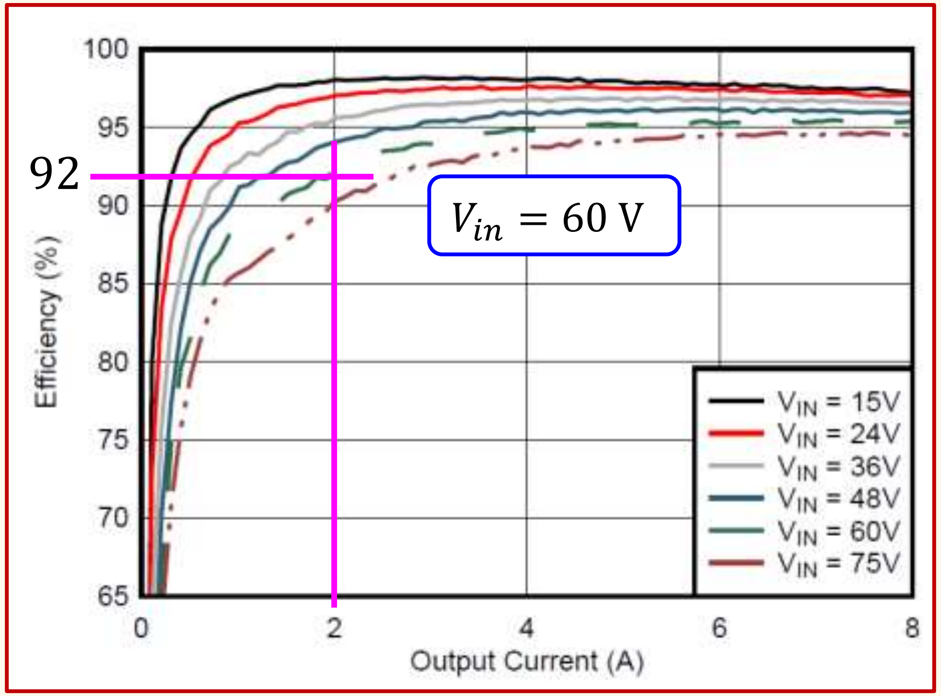
$L = 300\ \mu\text{H}$ $\quad r_L = 0.025\ \Omega$ $\quad R = 7.5\ \Omega$

$C = 20\ \mu\text{F}$ $\quad r_C = 0.400\ \Omega$ $\quad f_s = 100$ kHz (selected)

❖ Buck Controller IC (LM5146):

- Input Voltage Range: 5.5 V to 100 V
- Maximum Output Current: 20 A
- Control Mode: Voltage Mode
- Operating Temperature Range: -40 °C to 125 °C
- Reference Voltage: $V_{ref} = 0.8$ V
- PWM Ramp Amplitude: $V_{ramp} = \dfrac{V_S}{15} = \dfrac{60}{15} = 4$ V
- Error Amplifier Open-Loop DC Voltage Gain: $A_{OL} = 94$ dB (50118 V/V)
- Gain Bandwidth Product: $GBW = 6.5$ MHz
- Error Amplifier Open-Loop Compensation Pole: $f_{pe} = 129.7$ Hz

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
PWM CONTROL						
DC_{100kHz}	Maximum duty cycle	F_{SW} = 100 kHz, 6 V ≤ V_{VIN} ≤ 60 V	98	99		%
DC_{400kHz}		F_{SW} = 400 kHz, 6 V ≤ V_{VIN} ≤ 60 V	90	94		%
$V_{RAMP(min)}$	RAMP valley voltage (COMP at 0% duty cycle)			300		mV
k_{FF}	PWM feedforward gain (V_{IN} / V_{RAMP})	6 V ≤ V_{VIN} ≤ 100 V		15		V/V
ERROR AMPLIFIER						
V_{REF}	FB reference Voltage	FB connected to COMP	792	800	808	mV
AVOL	DC gain			94		dB
GBW	Unity gain bandwidth			6.5		MHz



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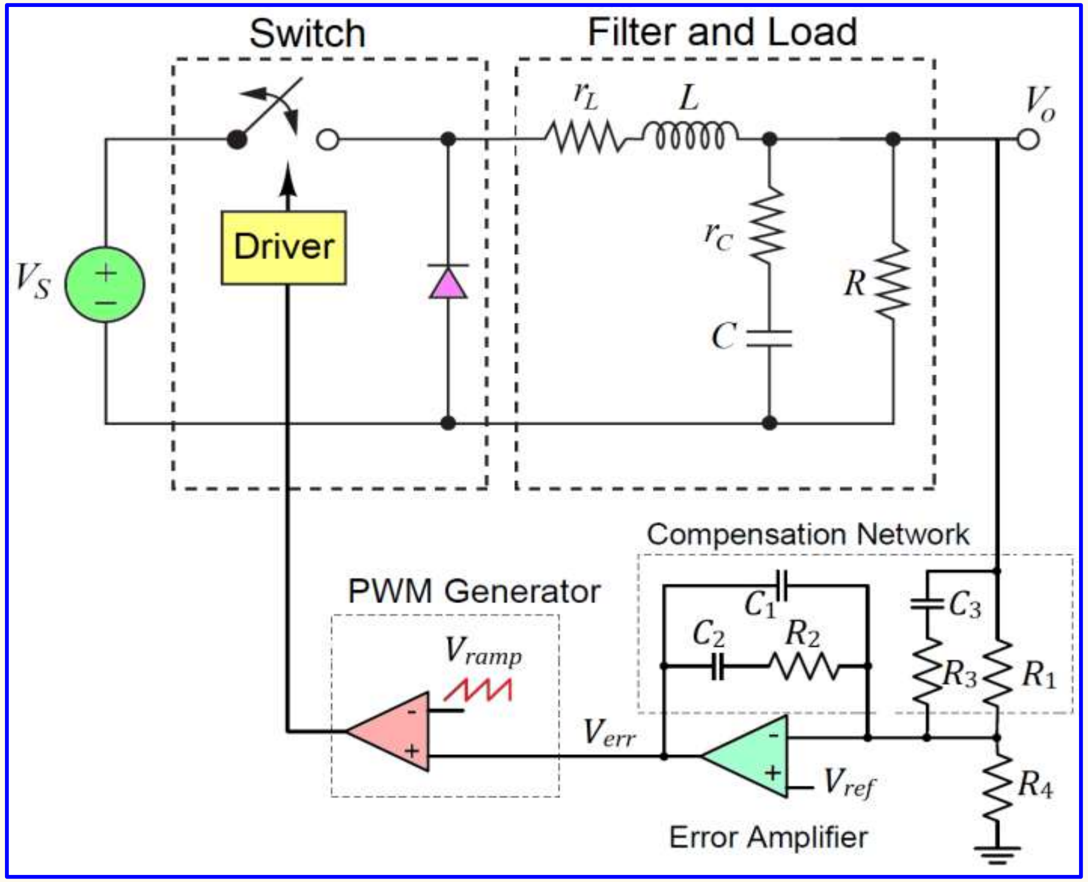

Type 3 Compensated Error Amplifier:

$$\boxed{G_3(s) = -\frac{(R_1 + R_3)(s + \omega_{z1})(s + \omega_{z2})}{R_1 R_3 C_1 s (s + \omega_{p2})(s + \omega_{p3})}}$$

$$G_3(s) = -\frac{R_1 + R_3}{R_1 R_3 C_1} \cdot \frac{\left(s + \dfrac{1}{(R_1 + R_3) C_3}\right)\left(s + \dfrac{1}{R_2 C_2}\right)}{s\left(s + \dfrac{1}{R_3 C_3}\right)\left(s + \dfrac{C_1 + C_2}{C_1 C_2 R_2}\right)}$$

Assuming:
$R_1 \gg R_3$
$C_2 \gg C_1$

$$G_3(s) \approx -\frac{1}{R_3 C_1} \cdot \frac{\left(s + \dfrac{1}{R_1 C_3}\right)\left(s + \dfrac{1}{R_2 C_2}\right)}{s\left(s + \dfrac{1}{R_3 C_3}\right)\left(s + \dfrac{1}{R_2 C_1}\right)}$$

❖ Compensator Values – *Pole Placement Method*:

- **Step 1:** *Resistor Divider Values* Select $R_4 = 10\ \mathrm{k}\Omega$

$$V_{ref} = \frac{R_4}{R_1 + R_4} V_o \Rightarrow R_1 = \left(\frac{V_o}{V_{ref}} - 1\right) R_4 = \left(\frac{15}{0.8} - 1\right) \cdot 10^4 \qquad \boxed{R_1 = 177.5\ \mathrm{k}\Omega}$$

- **Step 2:** *Resonant Frequency of the Filter*

$$f_{LC} = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{300 \cdot 10^{-6} \cdot 20 \cdot 10^{-6}}} = 2055\ \mathrm{Hz}$$

- **Step 3:** *First Zero Frequency*

Slightly smaller than f_{LC}

$$f_{z1} = 0.9 f_{LC} = 0.9 \cdot 2055 = 1850\ \mathrm{Hz}$$

$$f_{z1} = \frac{1}{2\pi R_1 C_3} \Rightarrow C_3 = \frac{1}{2\pi R_1 f_{z1}} = \frac{1}{2\pi \cdot 177.5 \cdot 10^3 \cdot 1850}$$

$$\boxed{C_3 = 484.7\ \mathrm{pF}}$$

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$$f_{z1} = \frac{1}{2\pi R_1 C_3} \Rightarrow C_3 = \frac{1}{2\pi R_1 f_{z1}} = \frac{1}{2\pi \cdot 177.5 \cdot 10^3 \cdot 1850}$$

$$\boxed{C_3 = 484.7\ \text{pF}}$$

- **Step 4:** *Second Pole Frequency*

Select the crossover frequency at $f_t = 10\ \text{kHz}$ (***one decade below*** $f_s = 100\ \text{kHz}$)   Set $f_{p2} = f_t = 10\ \text{kHz}$

$$f_{p2} = \frac{1}{2\pi R_3 C_3} \Rightarrow R_3 = \frac{1}{2\pi C_3 f_{p2}} = \frac{1}{2\pi \cdot 484.7 \cdot 10^{-12} \cdot 10^4} \qquad \boxed{R_3 = 32.84\ \text{k}\Omega}$$

- **Step 5:** *Required Compensator Gain at the Crossover Frequency*

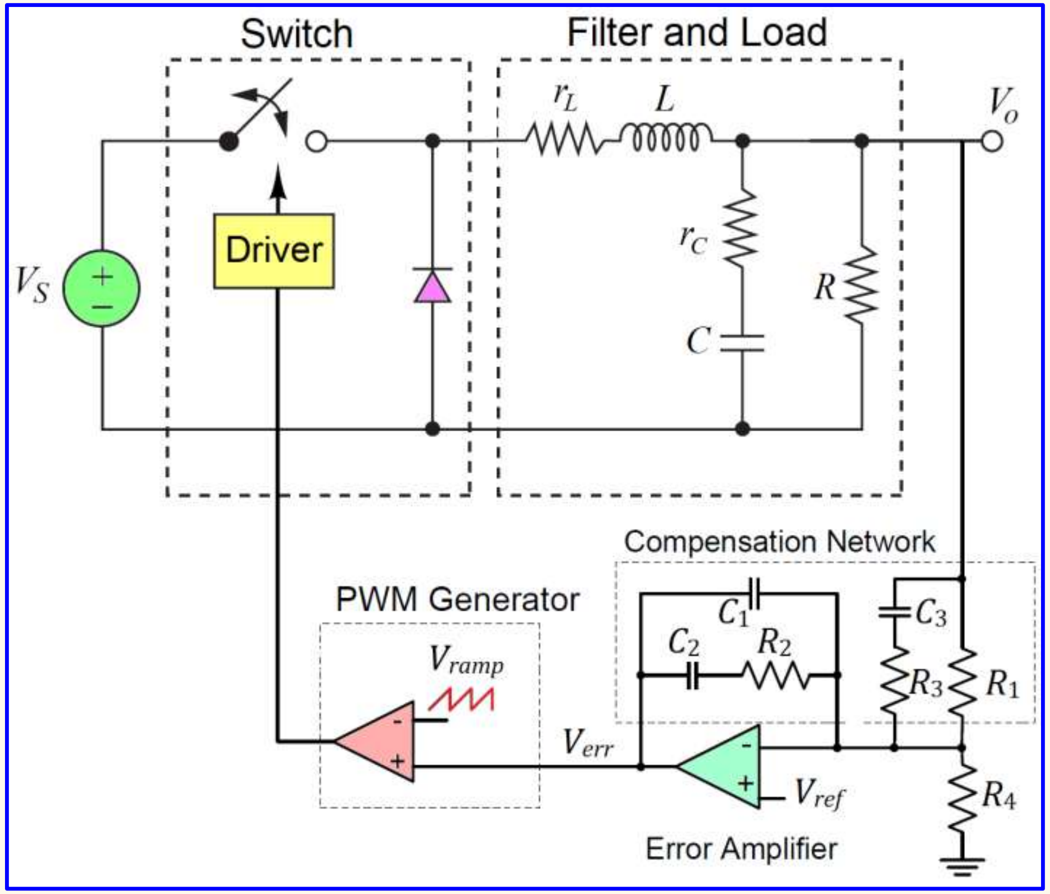
Modulator gain: $M = \dfrac{V_S}{V_{ramp}} = \dfrac{60}{4} = 15\ (23.52\ \text{dB})$

Filter gain: $$F(s) = \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right) LC s^2}$$

$$F(j\omega) = \frac{1 + j\omega r_C C}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega^2 + j\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}$$

$$|F(j\omega)| = \frac{\sqrt{1 + (\omega r_C C)^2}}{\sqrt{\left(1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega^2\right)^2 + \left(\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)\right)^2}}$$

$$|F(j\omega_t)| = 0.04636\ (-26.68\ \text{dB})$$



Type 3 Compensated Error Amplifier:

$$G_3(s) \approx -\frac{1}{R_3 C_1} \cdot \frac{\left(s + \frac{1}{R_1 C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{1}{R_2 C_1}\right)}$$

Loop gain at f_t must be unity:

$$K_{comp} \cdot M \cdot |F(j\omega_t)| = 1$$

$$K_{comp} = \frac{1}{M \cdot |F(j\omega_t)|}$$

$$K_{comp} = \frac{1}{15 \cdot 0.04636}$$

$$K_{comp} = 1.438\ (3.155\ \text{dB})$$

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❖ Compensator Values – *Pole Placement Method*:

- **Step 1:** *Resistor Divider Values* Select $R_4 = 10\ \text{k}\Omega$ $\boxed{R_1 = 177.5\ \text{k}\Omega}$
- **Step 2:** *Resonant Frequency of the Filter*

$$f_{LC} = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{300 \cdot 10^{-6} \cdot 20 \cdot 10^{-6}}} = 2055\ \text{Hz}$$

- **Step 3:** *First Zero Frequency*

Slightly smaller than f_{LC} $f_{z1} = 0.9 f_{LC} = 0.9 \cdot 2055 = 1850\ \text{Hz}$ $\boxed{C_3 = 484.7\ \text{pF}}$

- **Step 4:** *Second Pole Frequency*

Select the crossover frequency at $f_t = 10\ \text{kHz}$ (***one decade below*** $f_s = 100\ \text{kHz}$) Set $f_{p2} = f_t = 10\ \text{kHz}$

$$f_{p2} = \frac{1}{2\pi R_3 C_3} \Rightarrow R_3 = \frac{1}{2\pi C_3 f_{p2}} = \frac{1}{2\pi \cdot 484.7 \cdot 10^{-12} \cdot 10^4}$$ $\boxed{R_3 = 32.84\ \text{k}\Omega}$

- **Step 5:** *Required Compensator Gain at the Crossover Frequency*

$$K_{comp} = 1.438\ (3.155\ \text{dB})$$

- **Step 6:** *Compensator Gain Setting*

$$K_{comp} = \frac{R_2}{(R_1 || R_3)} \Rightarrow R_2 = K_{comp}(R_1 || R_3) = 1.438 \cdot \left(\frac{177.5 \cdot 10^3 \cdot 32.84 \cdot 10^3}{177.5 \cdot 10^3 + 32.84 \cdot 10^3}\right)$$ $\boxed{R_2 = 39.85\ \text{k}\Omega}$

- **Step 7:** *Second Zero Frequency*

Slightly smaller than f_{LC}

$f_{z2} = 0.9 f_{LC} = 0.9 \cdot 2055 = 1850\ \text{Hz}$

$$f_{z2} = \frac{1}{2\pi R_2 C_2} \Rightarrow C_2 = \frac{1}{2\pi R_2 f_{z2}} = \frac{1}{2\pi \cdot 39.85 \cdot 10^3 \cdot 1850}$$

$\boxed{C_2 = 2.159\ \text{nF}}$

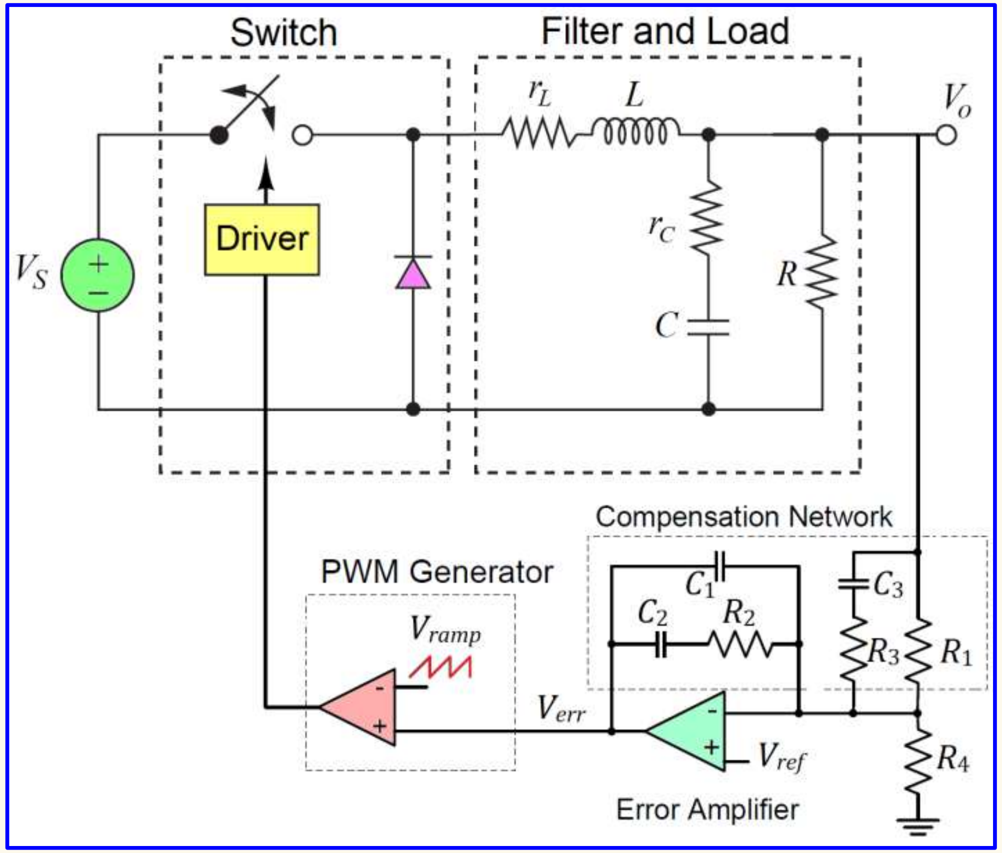
- **Step 8:** *Third Pole Frequency*

About one decade above $f_t = 10\ \text{kHz}$

$f_{p3} = 10 f_t = 10 \cdot 10^4 = 100\ \text{kHz}$

$$f_{p3} = \frac{1}{2\pi R_2 C_1} \Rightarrow C_1 = \frac{1}{2\pi R_2 f_{p3}} = \frac{1}{2\pi \cdot 39.85 \cdot 10^3 \cdot 10^5}$$

$\boxed{C_1 = 39.94\ \text{pF}}$



Type 3 Compensated Error Amplifier:

$$G_3(s) \approx -\frac{1}{R_3 C_1} \cdot \frac{\left(s + \frac{1}{R_1 C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{1}{R_2 C_1}\right)}$$

Summary:

$V_S = 60\ \text{V}$	$R = 7.5\ \Omega$	$R_1 = 177.5\ \text{k}\Omega$	$C_1 = 39.94\ \text{pF}$
$V_{ramp} = 4\ \text{V}$	$L = 300\ \mu\text{H}$	$R_2 = 39.85\ \text{k}\Omega$	$C_2 = 2.159\ \text{nF}$
	$C = 20\ \mu\text{F}$	$R_3 = 32.84\ \text{k}\Omega$	$C_3 = 484.7\ \text{pF}$
	$r_L = 0.025\ \Omega$	$R_4 = 10\ \text{k}\Omega$	
	$r_C = 0.400\ \Omega$		

❖ Total Loop Transfer Function:

$$L(s) = M(s) F(s) G_3(s)$$

$$L(s) = \frac{V_S}{V_{ramp}} \cdot \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right)LCs^2} \cdot \frac{1}{R_3 C_1} \cdot \frac{\left(s + \frac{1}{R_1 C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{1}{R_2 C_1}\right)}$$

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PART 2C:

Simulations

MATLAB/Simulink

Can Bijles

Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\ \mathrm{V}$
- Average output voltage $V_o = 15\ \mathrm{V}$
- Average output current $I_o = 2\ \mathrm{A}$

Solution: ❑ Simulations – MATLAB Script

Transfer Functions:

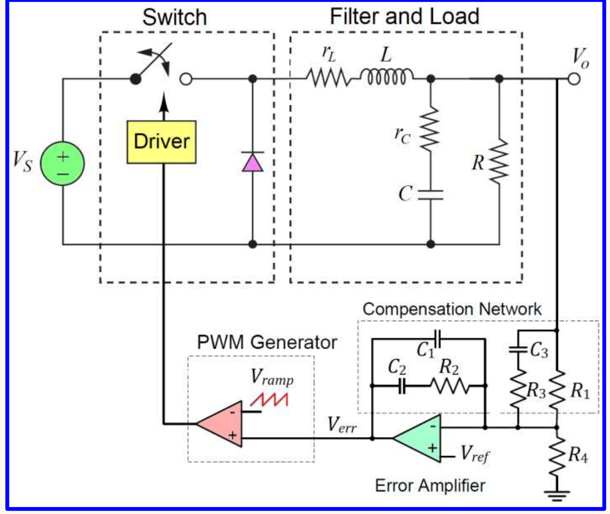
- Modulator: $M(s) = \dfrac{V_S}{V_{ramp}}$
- Filter & Load: $F(s) = \dfrac{1 + r_C C s}{1 + \dfrac{r_L}{R} + \left(\dfrac{L}{R} + (r_L + r_C)C + \dfrac{r_L r_C C}{R}\right)s + \left(1 + \dfrac{r_C}{R}\right)LCs^2}$
- Compensator: $G_3(s) = \dfrac{1}{R_3 C_1} \cdot \dfrac{\left(s + \dfrac{1}{R_1 C_3}\right)\left(s + \dfrac{1}{R_2 C_2}\right)}{s\left(s + \dfrac{1}{R_3 C_3}\right)\left(s + \dfrac{1}{R_2 C_1}\right)}$
- Loop: $L(s) = M(s)F(s)G_3(s)$

```matlab
s = tf('s'); % Define a Laplace parameter

%**SPECIFICATIONS**
Vs = 60; Vo = 15; % Input and output DC voltage values

%**SELECTIONS**
f_s = 1e5; % Switching frequency

%**BUCK CONTROLLER IC: LM5146**
Vref = 0.8; % Reference voltage from Buck Controller LM5146
Vramp = Vs/15; %Ramp voltage amplitude from Buck Controller LM5146
A_DC = 50119; % Error amplifier open-loop DC gain
f_pe = 129.7; % Error amplifier open-loop pole

%**COMPONENTS**
R = 7.5; L = 300e-6; C = 20e-6; % Buck converter output filter values
r_C = 0.400; % ESR of the capacitor C
r_L = 0.025; % ESR of the inductor L

%**TRANSFER FUNCTIONS**
M = Vs/Vramp; % PWM modulator transfer function
F = (1+r_C*C*s)/(1+r_L/R + (L/R+(r_C+r_L)*C+r_C*r_L*C/R)*s + (1+r_C/R)*L*C*s^2); % Filter and Load

%**BREAK FREQUENCIES**
f_LC = 1/(2*pi*sqrt(L*C)); % Filter's resonant frequency
f_z1 = 0.9*f_LC; % First zero frequency slightly lower than f_LC
f_t = 0.1*f_s; % Crossover frequency (selected value one decade below f_s)

%**CONTROLLER COMPONENT VALUES**
R4 = 1e4; % Bottom resistor for reference voltage divider (selected)
R1 = (Vo/Vref - 1)*R4; % Top resistor for reference voltage divider
C3 = 1/(2*pi*f_z1*R1);
R3 = 1/(2*pi*f_t*C3);

G_MF = M*F; % Loop transfer function Modulator + Filter
K_comp = 1.438; % Required compensator gain at the crossover frequency

R2 = K_comp*(R1^-1+R3^-1)^-1;
C2 = 1/(2*pi*0.9*f_LC*R2);
C1 = 1/(2*pi*10*f_t*R2);

%**COMPENSATOR TRANSFER FUNCTION**
G_OL = -A_DC/(1+s/(2*pi*f_pe)); % Error amplifier transfer
beta = (R1*R3*C1*s*(s+(C1+C2)/(C1*C2*R2))*(s+1/(R3*C3)))/((R1+R3)*(s+1/(R2*C2))*(s+1/((R1+R3)*C3)));
G_comp = G_OL/(1+G_OL*beta); % Compensator transfer
G_loop = M*F*G_comp;
```

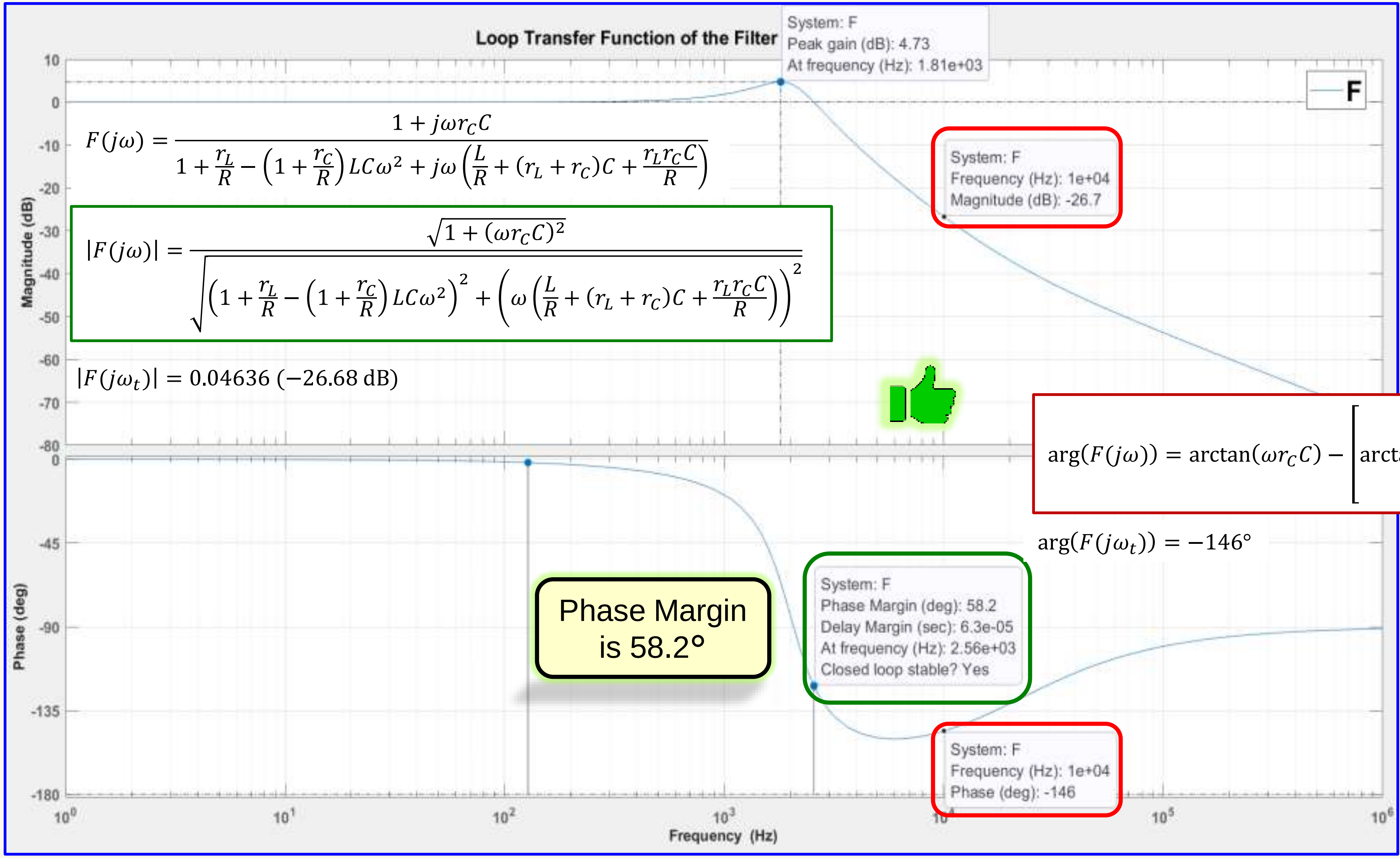
Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\text{ V}$
- Average output voltage $V_o = 15\text{ V}$
- Average output current $I_o = 2\text{ A}$

Summary: $V_S = 60\text{ V}$, $V_{ramp} = 4\text{ V}$, $R = 7.5\ \Omega$, $L = 300\ \mu\text{H}$, $C = 20\ \mu\text{F}$, $r_L = 0.025\ \Omega$, $r_C = 0.400\ \Omega$, $R_1 = 177.5\text{ k}\Omega$, $R_2 = 39.85\text{ k}\Omega$, $R_3 = 32.84\text{ k}\Omega$, $R_4 = 10\text{ k}\Omega$, $C_1 = 39.94\text{ pF}$, $C_2 = 2.159\text{ nF}$, $C_3 = 484.7\text{ pF}$



Solution: ❑ *Simulations – Loop Transfer Function of the Filter*



$$F(j\omega) = \frac{1 + j\omega r_C C}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right)LC\omega^2 + j\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}$$

$$|F(j\omega)| = \frac{\sqrt{1 + (\omega r_C C)^2}}{\sqrt{\left(1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right)LC\omega^2\right)^2 + \left(\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)\right)^2}}$$

$$|F(j\omega_t)| = 0.04636\ (-26.68\text{ dB})$$

$$\arg(F(j\omega)) = \arctan(\omega r_C C) - \left[\arctan\left(\frac{\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right)LC\omega^2}\right) + 180°\right]$$

$$\arg(F(j\omega_t)) = -146°$$

Phase Margin is 58.2°

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Objective: Design a buck converter circuit with:

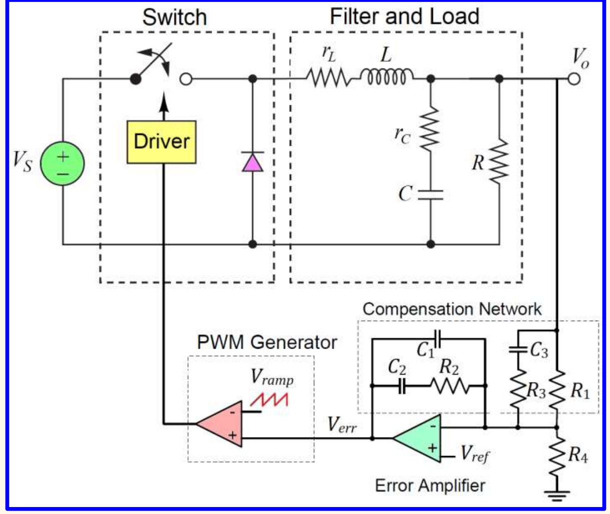
- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$

Summary: $V_S = 60\ \text{V}$, $V_{ramp} = 4\ \text{V}$

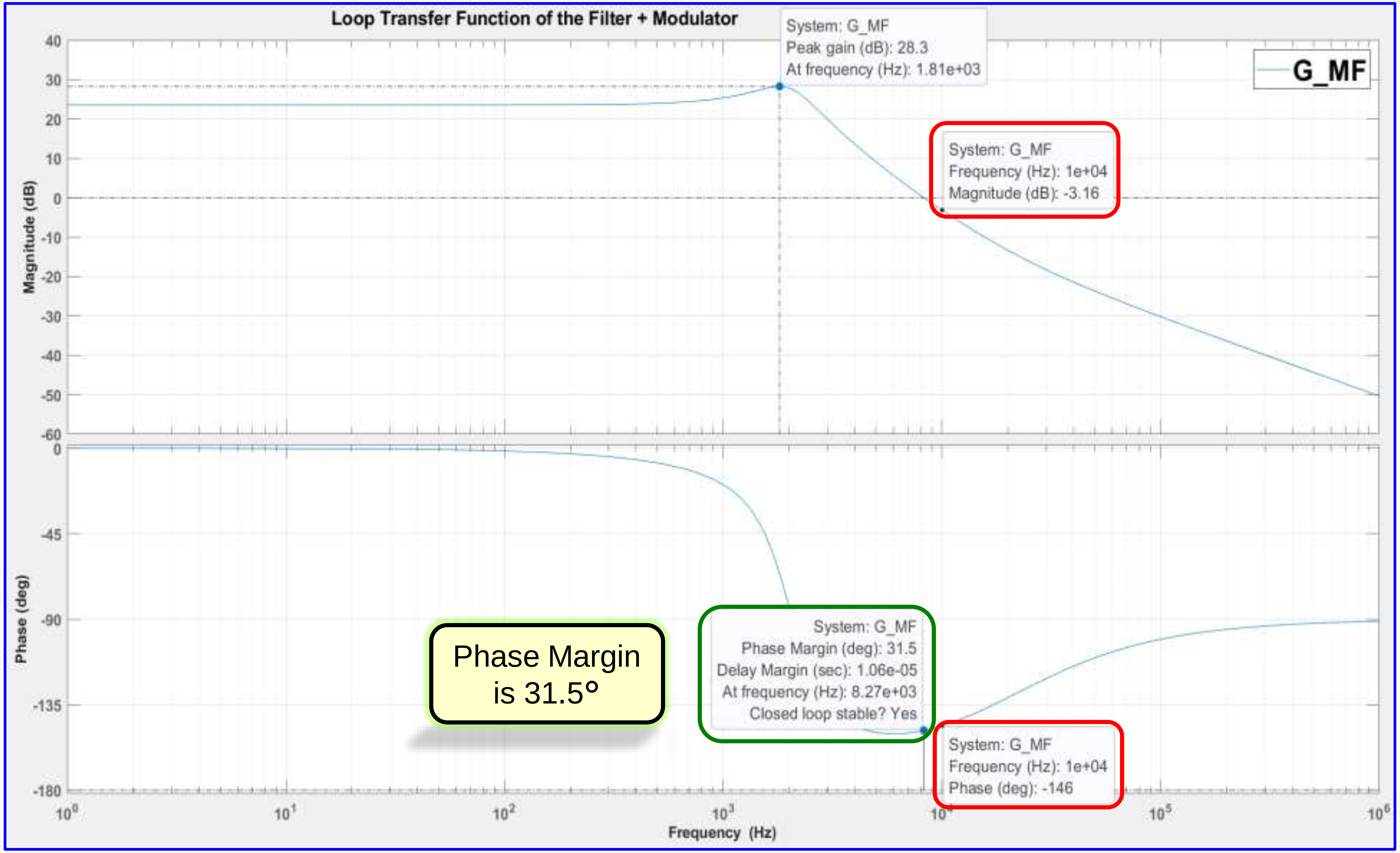
$R = 7.5\ \Omega$, $L = 300\ \mu\text{H}$, $C = 20\ \mu\text{F}$, $r_L = 0.025\ \Omega$, $r_C = 0.400\ \Omega$

$R_1 = 177.5\ \text{k}\Omega$, $R_2 = 39.85\ \text{k}\Omega$, $R_3 = 32.84\ \text{k}\Omega$, $R_4 = 10\ \text{k}\Omega$

$C_1 = 39.94\ \text{pF}$, $C_2 = 2.159\ \text{nF}$, $C_3 = 484.7\ \text{pF}$



Solution: ❑ *Simulations – Loop Transfer Function of the Filter + Modulator*



Objective: Design a buck converter circuit with:

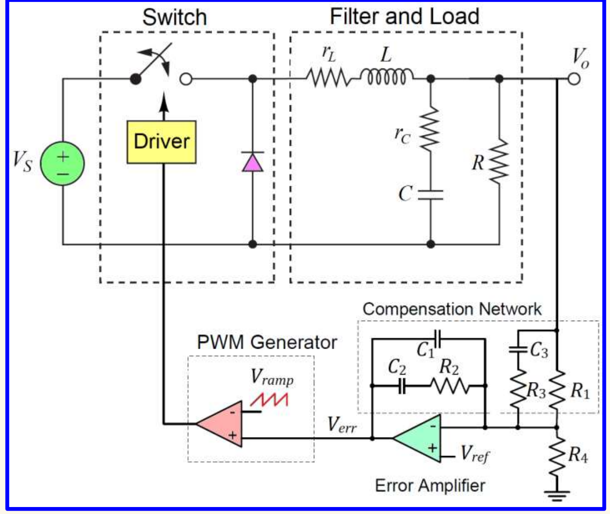
- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$

Summary: $V_S = 60\ \text{V}$, $V_{ramp} = 4\ \text{V}$

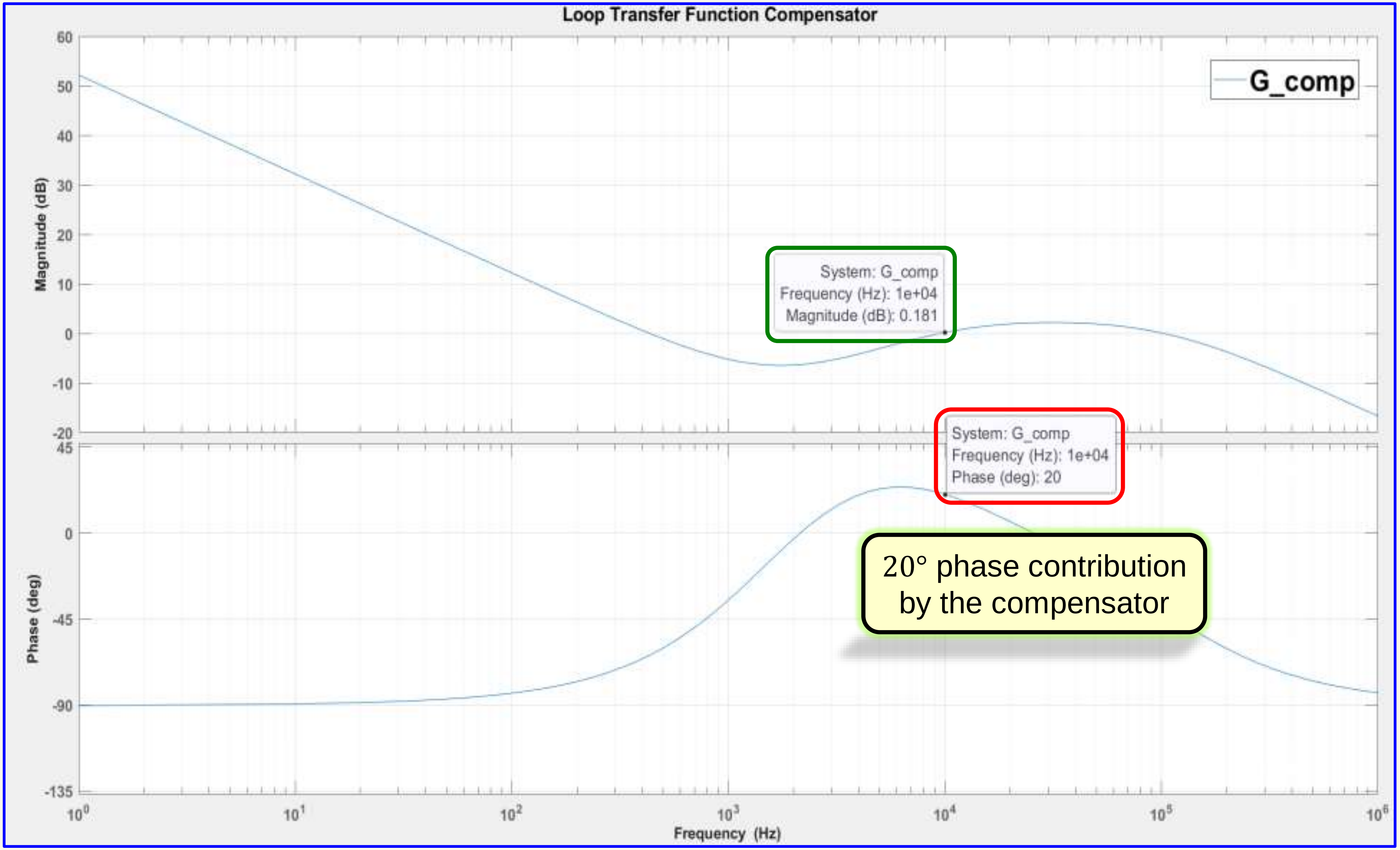
$R = 7.5\ \Omega$, $L = 300\ \mu\text{H}$, $C = 20\ \mu\text{F}$, $r_L = 0.025\ \Omega$, $r_C = 0.400\ \Omega$

$R_1 = 177.5\ \text{k}\Omega$, $R_2 = 39.85\ \text{k}\Omega$, $R_3 = 32.84\ \text{k}\Omega$, $R_4 = 10\ \text{k}\Omega$

$C_1 = 39.94\ \text{pF}$, $C_2 = 2.159\ \text{nF}$, $C_3 = 484.7\ \text{pF}$



Solution: ❑ *Simulations – Loop Transfer Function Compensator*



Objective: Design a buck converter circuit with:

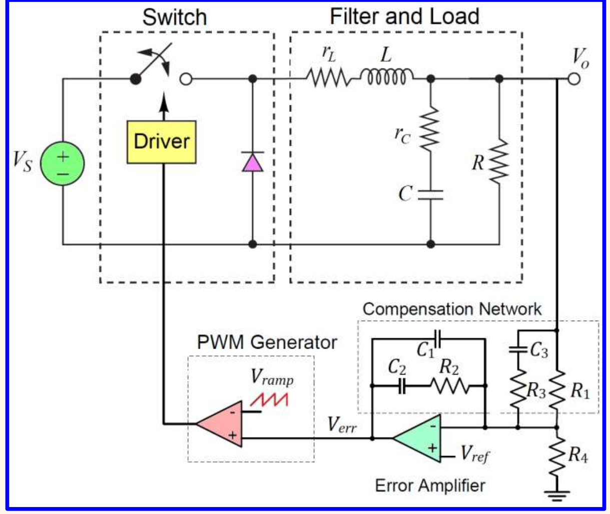
- DC input voltage $V_S = 60$ V
- Average output voltage $V_o = 15$ V
- Average output current $I_o = 2$ A

Summary: $V_S = 60$ V, $V_{ramp} = 4$ V

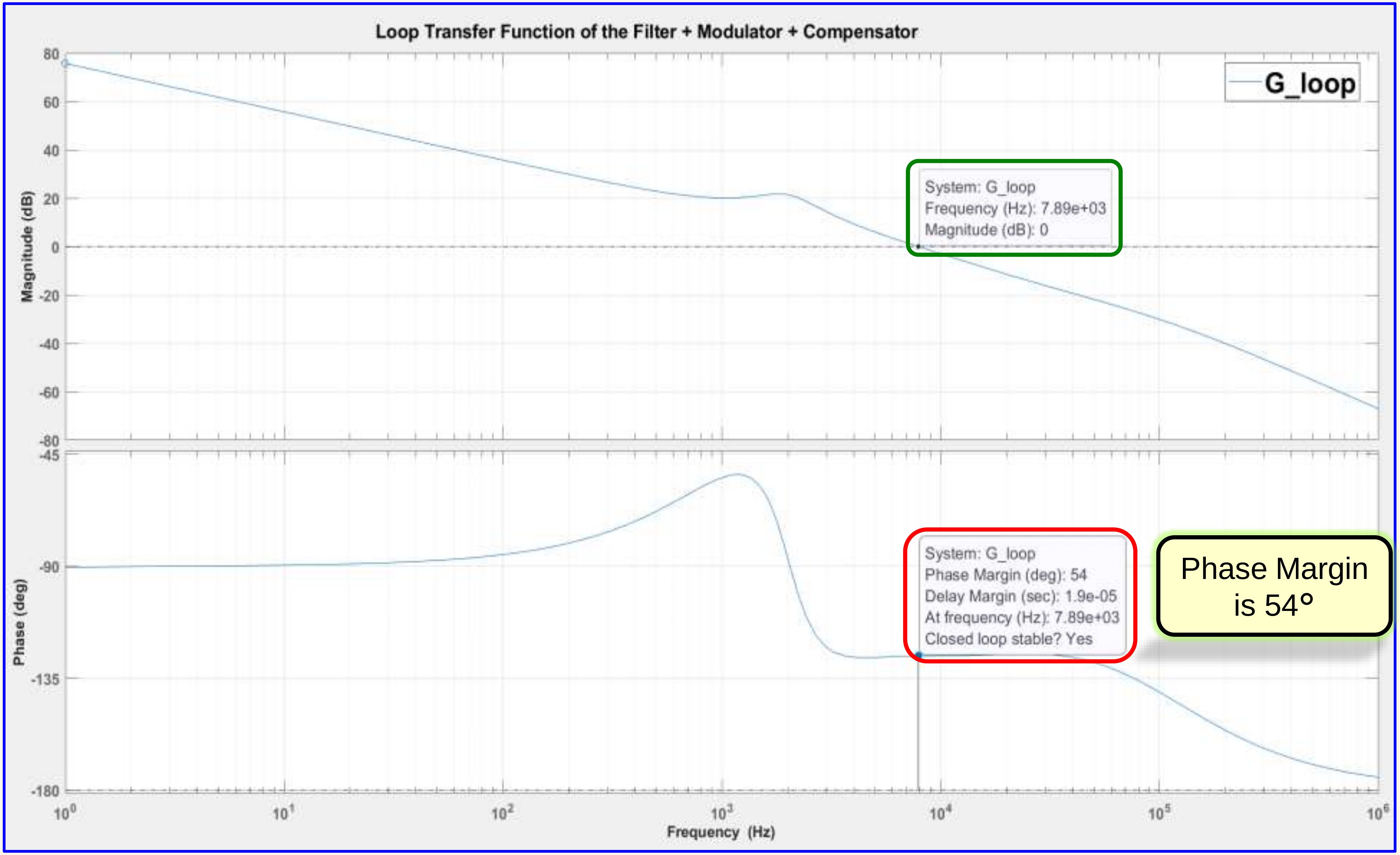
$R = 7.5\ \Omega$, $L = 300\ \mu$H, $C = 20\ \mu$F, $r_L = 0.025\ \Omega$, $r_C = 0.400\ \Omega$

$R_1 = 177.5$ kΩ, $R_2 = 39.85$ kΩ, $R_3 = 32.84$ kΩ, $R_4 = 10$ kΩ

$C_1 = 39.94$ pF, $C_2 = 2.159$ nF, $C_3 = 484.7$ pF



Solution: ❑ *Simulations – Loop Transfer Function Filter + Modulator + Compensator*



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PART 2C:

Simulations

TINA-TI SPICE

Objective: Design a buck converter circuit with:

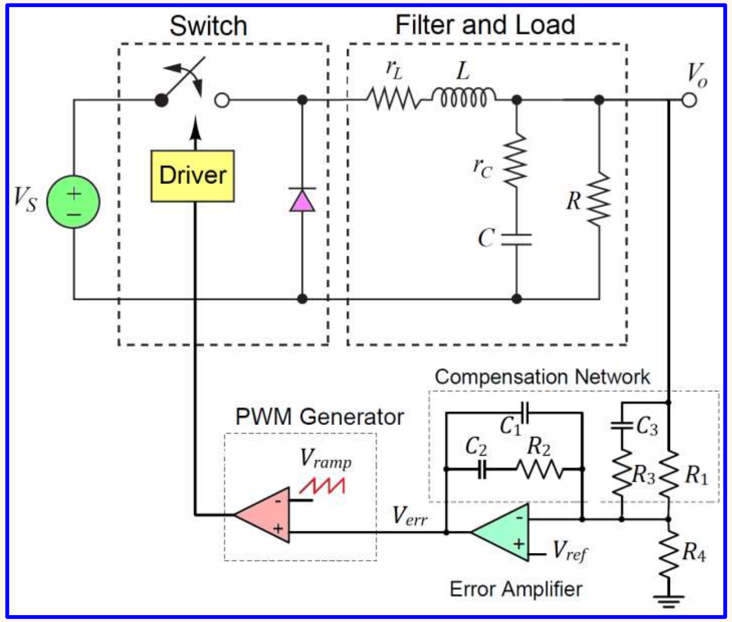
- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$

Summary: $V_S = 60\ \text{V}$, $V_{ramp} = 4\ \text{V}$

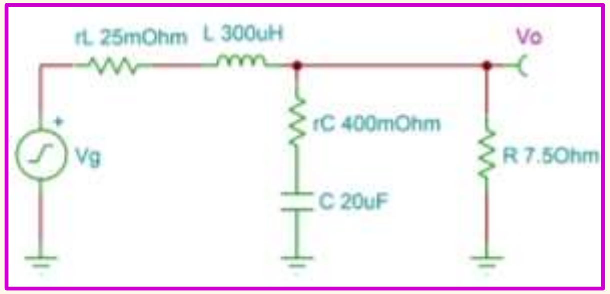
$R = 7.5\ \Omega$, $L = 300\ \mu\text{H}$, $C = 20\ \mu\text{F}$, $r_L = 0.025\ \Omega$, $r_C = 0.400\ \Omega$

$R_1 = 177.5\ \text{k}\Omega$, $R_2 = 39.85\ \text{k}\Omega$, $R_3 = 32.84\ \text{k}\Omega$, $R_4 = 10\ \text{k}\Omega$

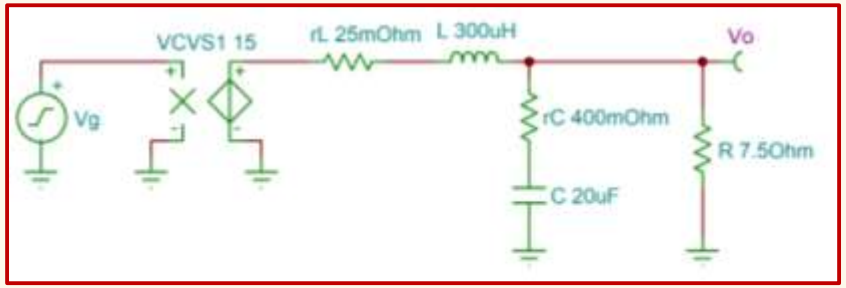
$C_1 = 39.94\ \text{pF}$, $C_2 = 2.159\ \text{nF}$, $C_3 = 484.7\ \text{pF}$



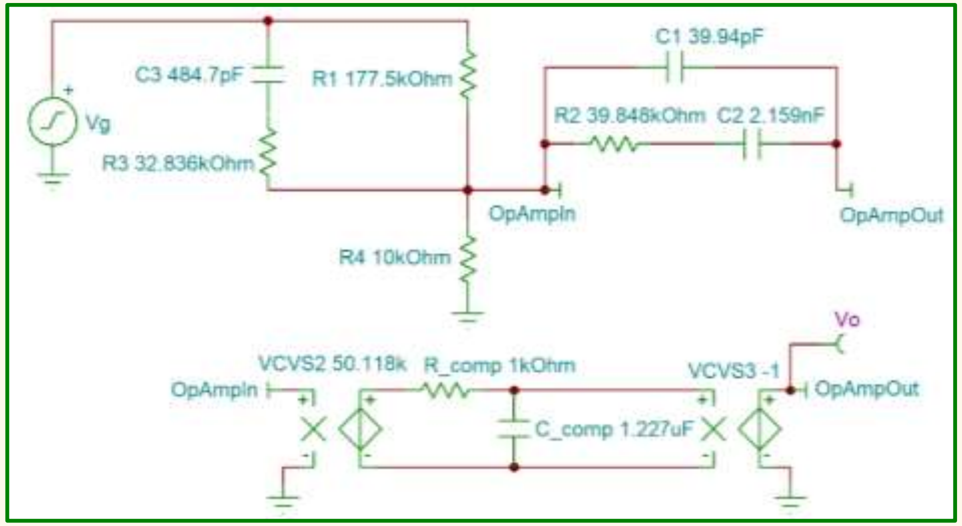
Solution: ❑ *Simulations – Circuits*



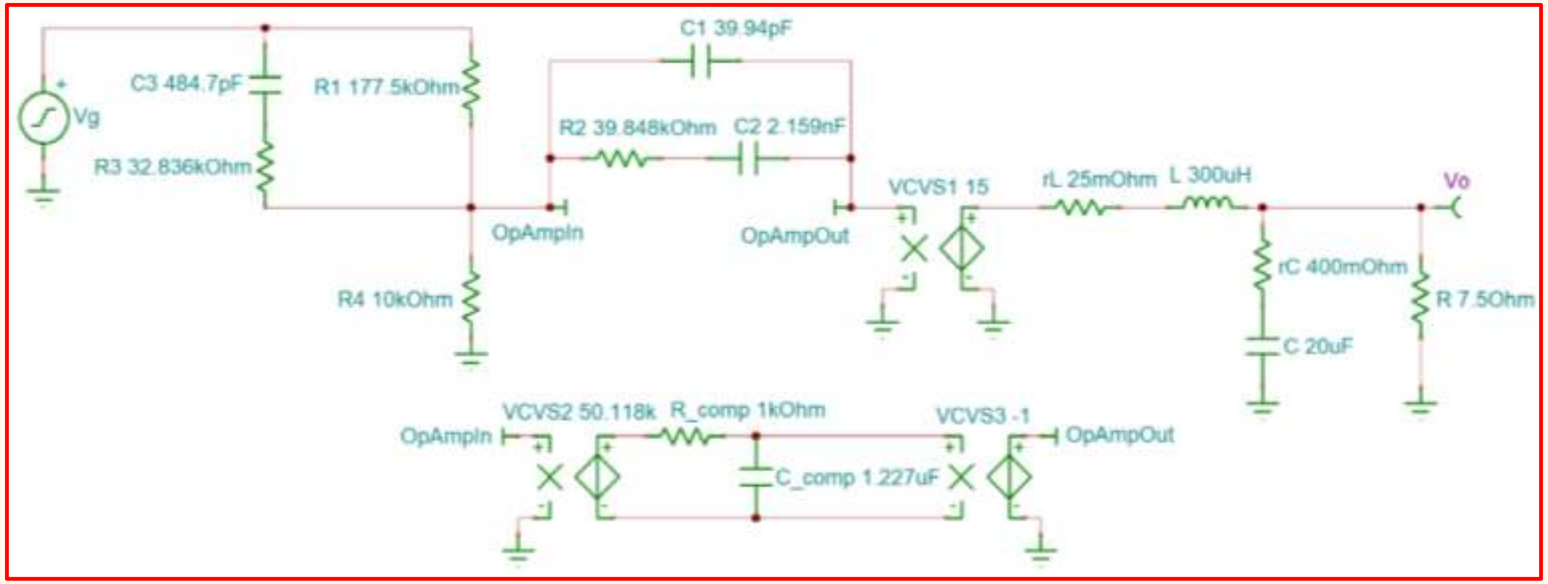
Filter



Filter & Modulator



Compensator



Filter & Modulator & Compensator

- Error Amplifier Open-Loop DC Voltage Gain: $A_{OL} = 94\ \text{dB}\ (50118\ \text{V/V})$
- Error Amplifier Open-Loop Compensation Pole: $f_{pe} = 129.7\ \text{Hz}$

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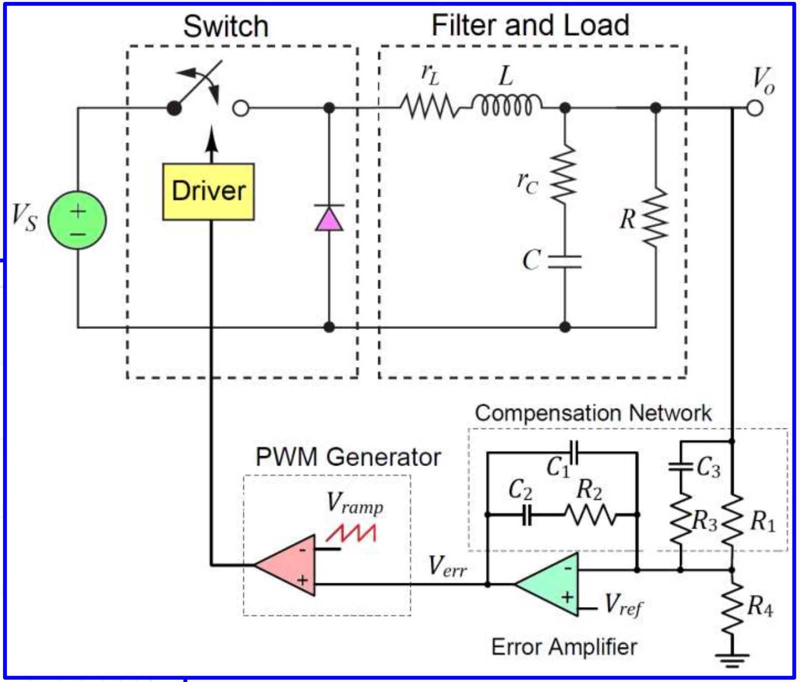
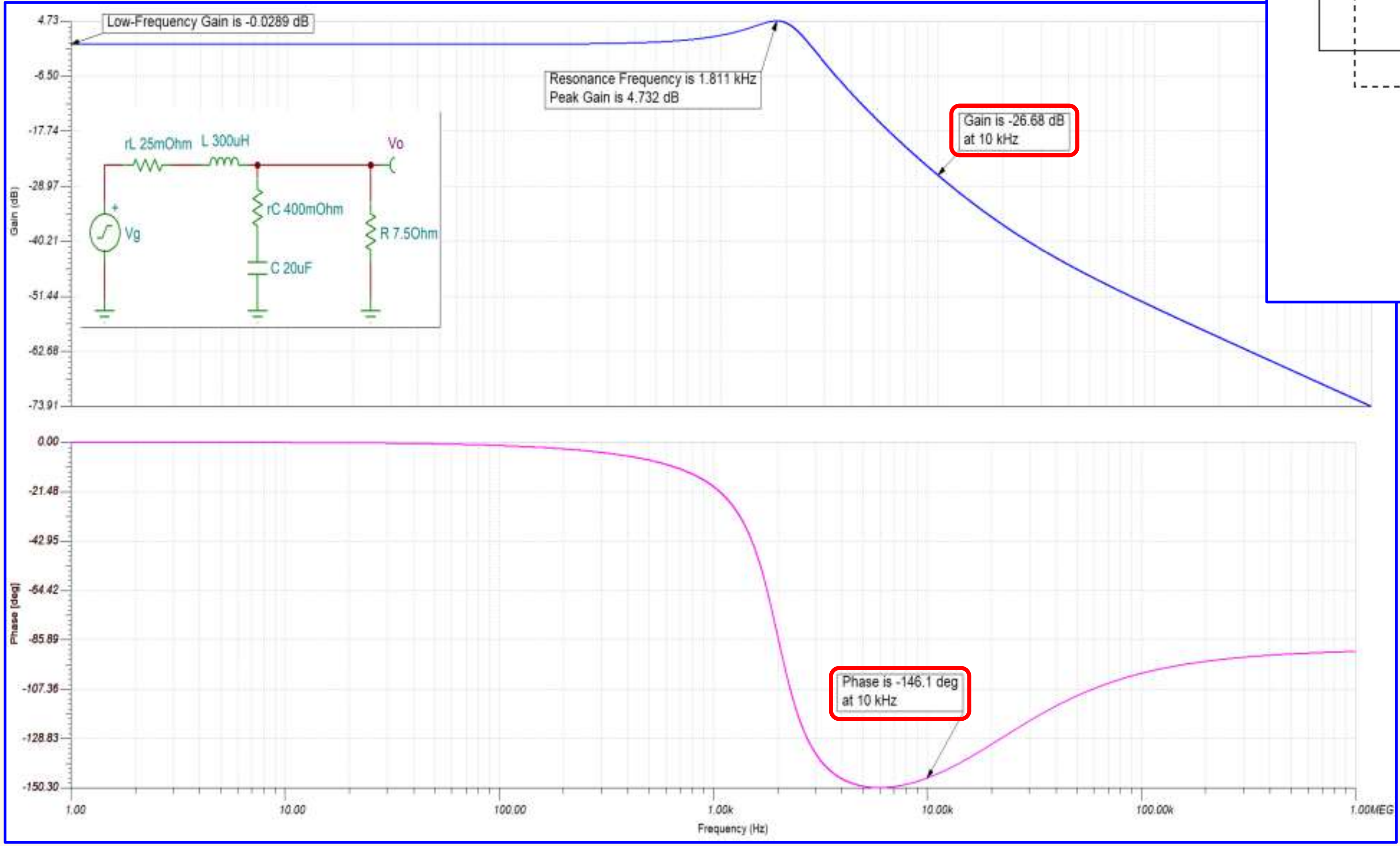
Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\ \mathrm{V}$
- Average output voltage $V_o = 15\ \mathrm{V}$
- Average output current $I_o = 2\ \mathrm{A}$



Solution: ❑ *Simulations – Loop Transfer Function of the Filter*



Summary:

$V_S = 60\ \mathrm{V}$

$V_{ramp} = 4\ \mathrm{V}$

$R = 7.5\ \Omega$

$L = 300\ \mu\mathrm{H}$

$C = 20\ \mu\mathrm{F}$

$r_L = 0.025\ \Omega$

$r_C = 0.400\ \Omega$

$R_1 = 177.5\ \mathrm{k}\Omega$ $\quad C_1 = 39.94\ \mathrm{pF}$

$R_2 = 39.85\ \mathrm{k}\Omega$ $\quad C_2 = 2.159\ \mathrm{nF}$

$R_3 = 32.84\ \mathrm{k}\Omega$ $\quad C_3 = 484.7\ \mathrm{pF}$

$R_4 = 10\ \mathrm{k}\Omega$

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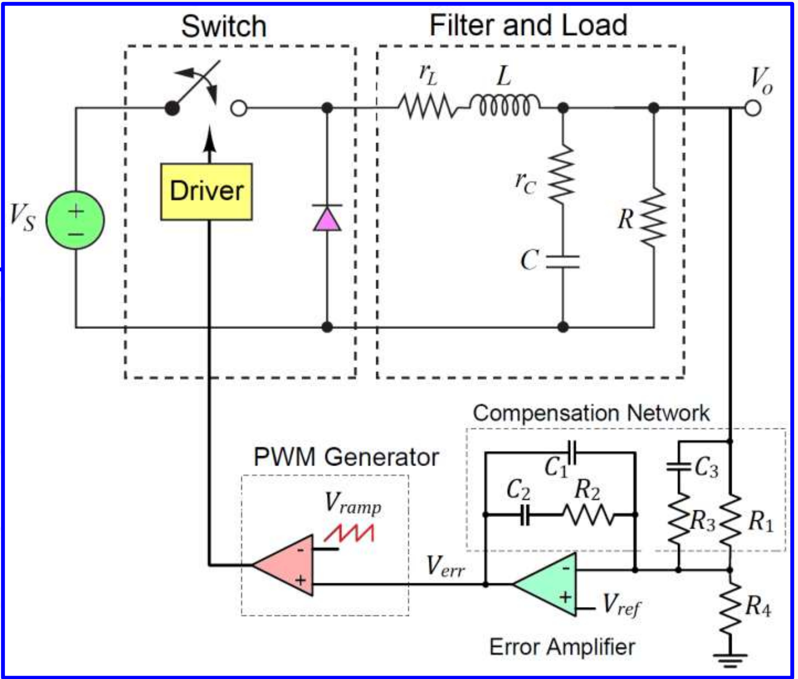
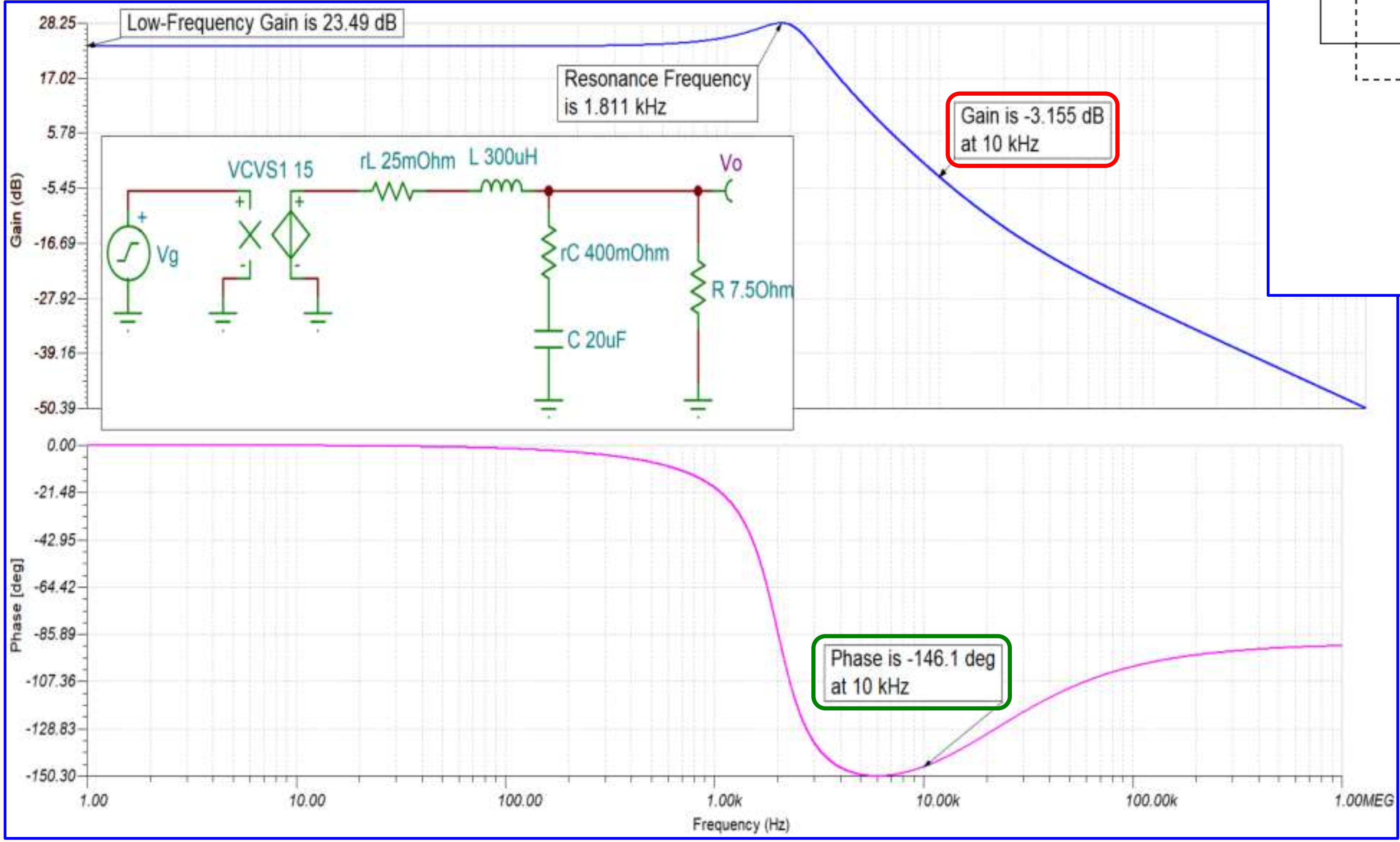
Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$

Solution: ❑ *Simulations – Loop Transfer Function of the Filter + Modulator*





Summary:

$V_S = 60\ \text{V}$

$V_{ramp} = 4\ \text{V}$

$R = 7.5\ \Omega$

$L = 300\ \mu\text{H}$

$C = 20\ \mu\text{F}$

$r_L = 0.025\ \Omega$

$r_C = 0.400\ \Omega$

$R_1 = 177.5\ \text{k}\Omega$

$R_2 = 39.85\ \text{k}\Omega$

$R_3 = 32.84\ \text{k}\Omega$

$R_4 = 10\ \text{k}\Omega$

$C_1 = 39.94\ \text{pF}$

$C_2 = 2.159\ \text{nF}$

$C_3 = 484.7\ \text{pF}$

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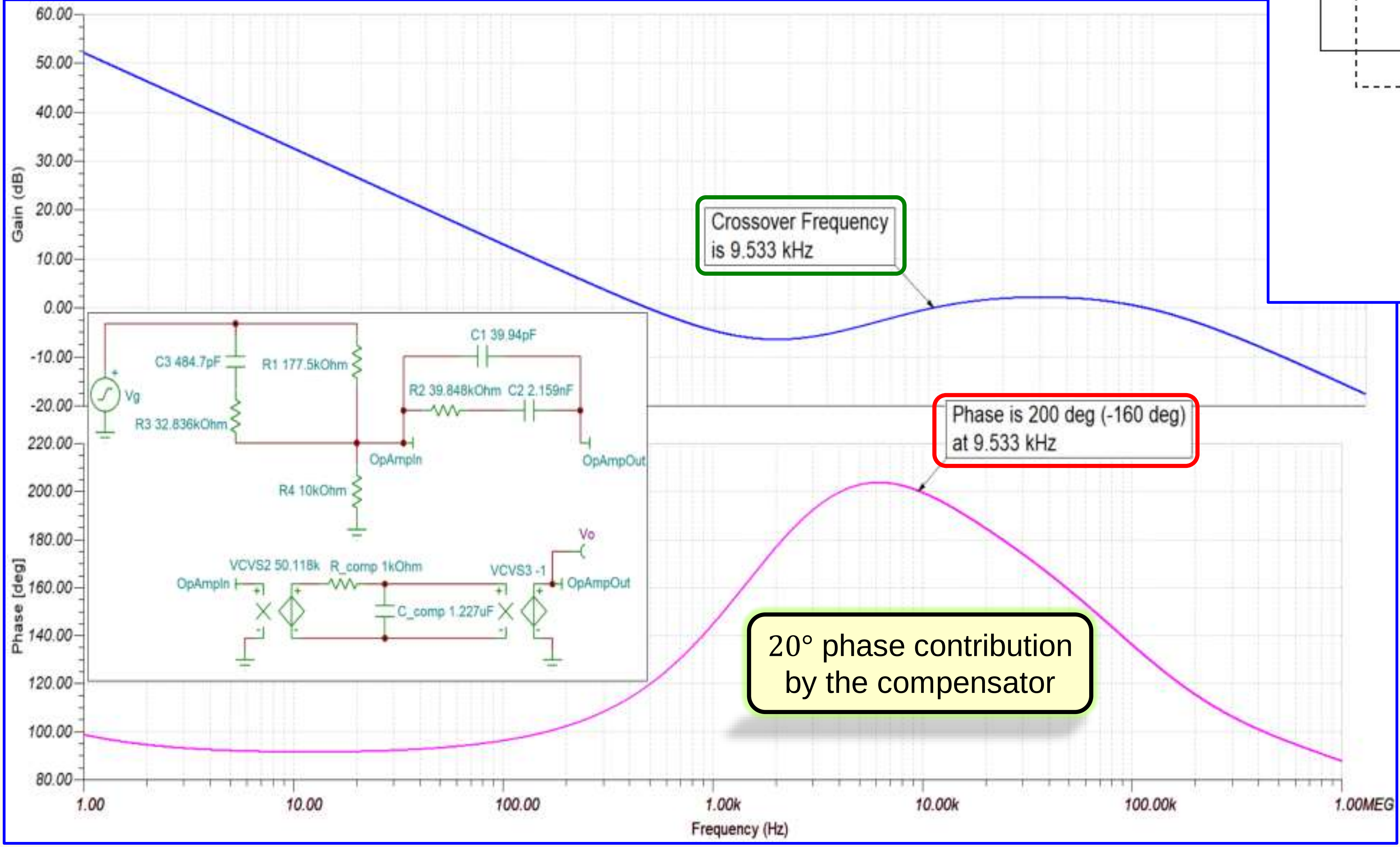
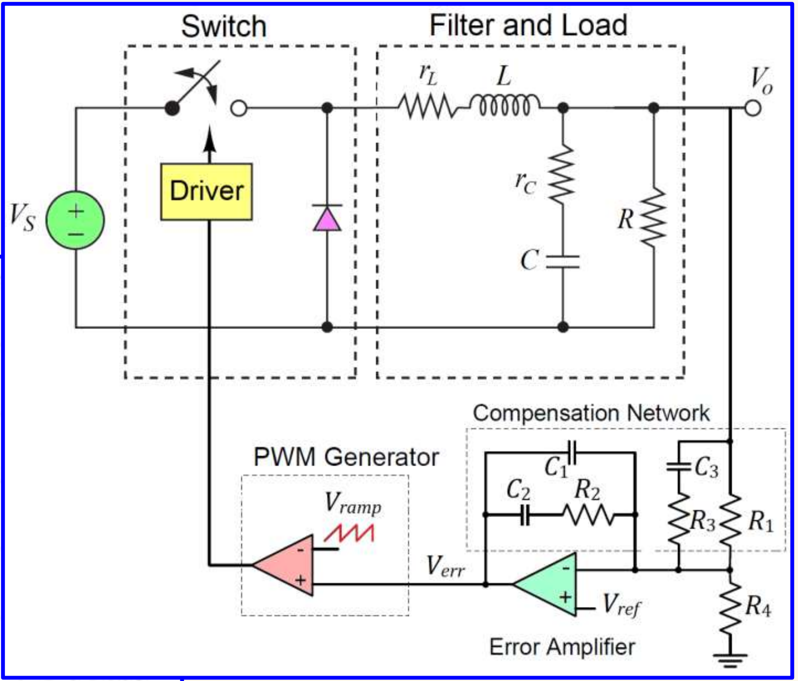
Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\ \mathrm{V}$
- Average output voltage $V_o = 15\ \mathrm{V}$
- Average output current $I_o = 2\ \mathrm{A}$

Solution: ❑ *Simulations – Loop Transfer Function Compensator*





Summary:

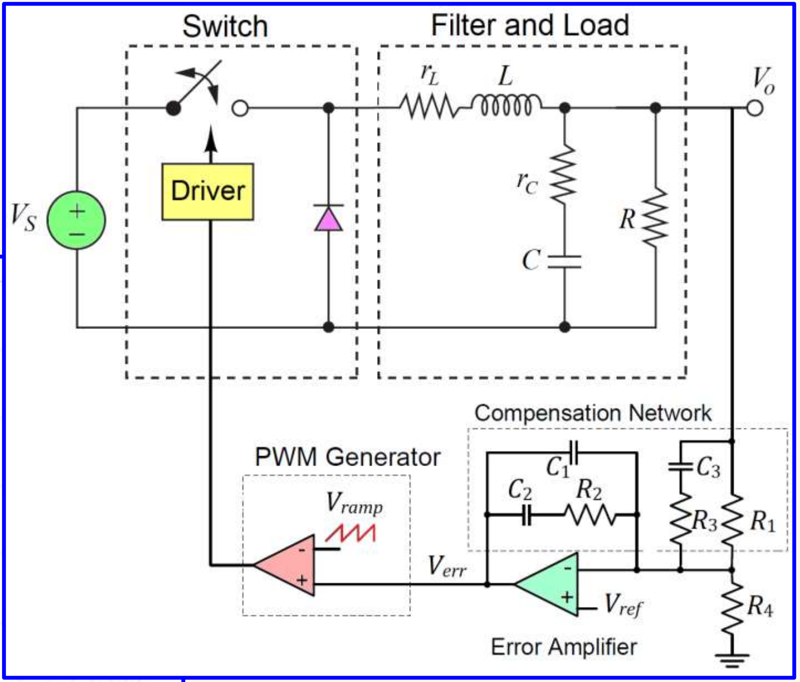
$V_S = 60\ \mathrm{V}$	$R = 7.5\ \Omega$
$V_{ramp} = 4\ \mathrm{V}$	$L = 300\ \mu\mathrm{H}$
	$C = 20\ \mu\mathrm{F}$
	$r_L = 0.025\ \Omega$
	$r_C = 0.400\ \Omega$
$R_1 = 177.5\ \mathrm{k}\Omega$	$C_1 = 39.94\ \mathrm{pF}$
$R_2 = 39.85\ \mathrm{k}\Omega$	$C_2 = 2.159\ \mathrm{nF}$
$R_3 = 32.84\ \mathrm{k}\Omega$	$C_3 = 484.7\ \mathrm{pF}$
$R_4 = 10\ \mathrm{k}\Omega$	

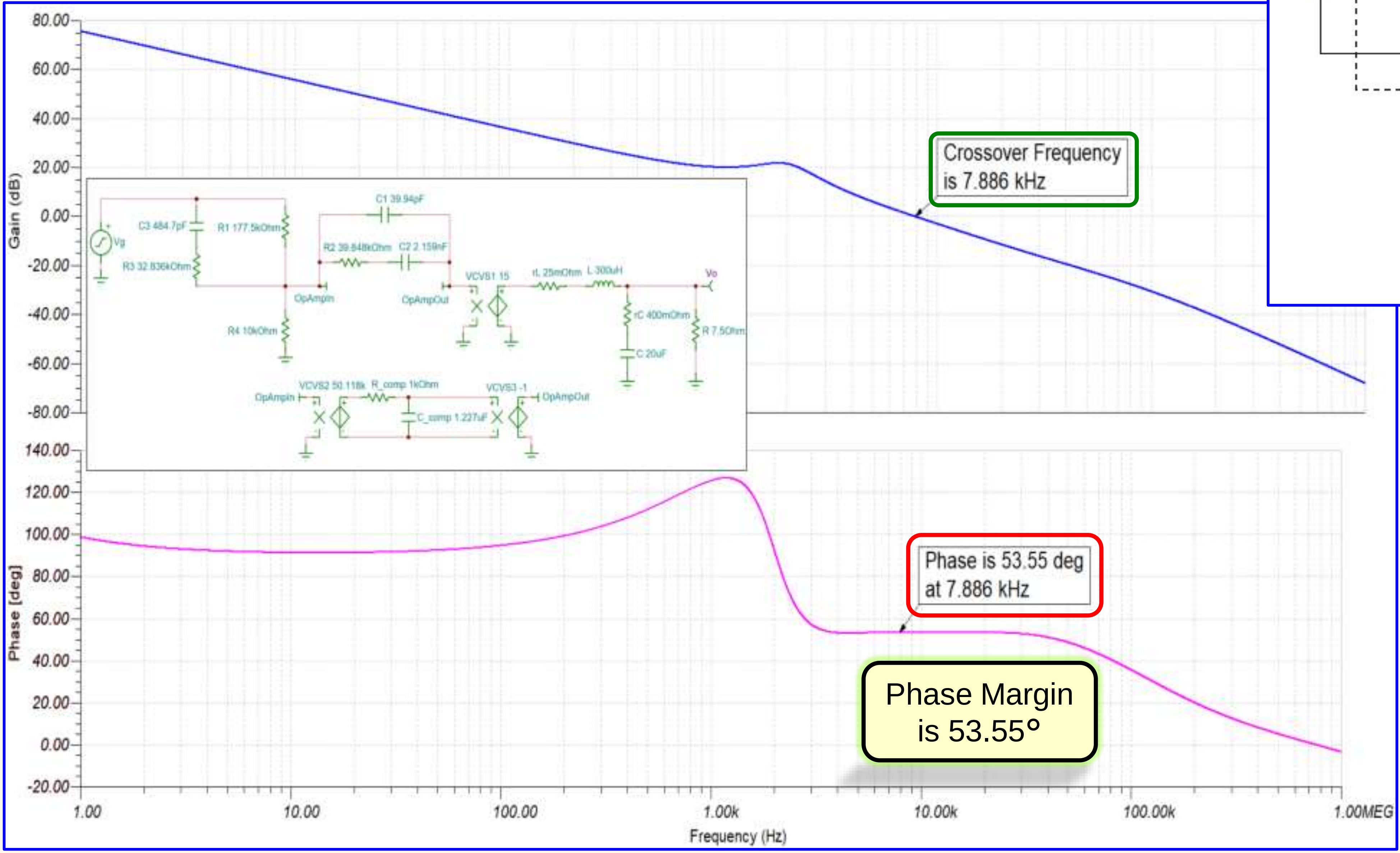
CanBijles

Objective: Design a buck converter circuit with:

- DC input voltage $V_S = 60\ \mathrm{V}$
- Average output voltage $V_o = 15\ \mathrm{V}$
- Average output current $I_o = 2\ \mathrm{A}$

Solution: ❑ *Simulations – Loop Transfer Function Filter + Modulator + Compensator*





Summary:

$V_S = 60\ \mathrm{V}$

$V_{ramp} = 4\ \mathrm{V}$

$R = 7.5\ \Omega$

$L = 300\ \mu\mathrm{H}$

$C = 20\ \mu\mathrm{F}$

$r_L = 0.025\ \Omega$

$r_C = 0.400\ \Omega$

$R_1 = 177.5\ \mathrm{k\Omega}$

$R_2 = 39.85\ \mathrm{k\Omega}$

$R_3 = 32.84\ \mathrm{k\Omega}$

$R_4 = 10\ \mathrm{k\Omega}$

$C_1 = 39.94\ \mathrm{pF}$

$C_2 = 2.159\ \mathrm{nF}$

$C_3 = 484.7\ \mathrm{pF}$

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